rph8-g15q — formula report

489 inline formulas, 81 display equations, 0 tables, 12 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_EQ0018 (18)	005	■ 0.837 ● N|+1	`\begin{aligned} \text { SWAP } L_{1} \text { SWAP } & =L_{2} \\ \left\{X \otimes X, L_{1}\right\} & =0 \end{aligned}`	$\begin{aligned} \text { SWAP } L_{1} \text { SWAP } & =L_{2} \\ \left\{X \otimes X, L_{1}\right\} & =0 \end{aligned}$	$\begin{aligned} \texttt{SWAP}\, L_{1}\, \texttt{SWAP} &= L_{2}, \\ \{X \otimes X, L_{1}\} &= 0 \end{aligned}$
rph8-g15q_EQ0065 (D1)	019	■ 0.848 ● W|+0	`\begin{aligned} d\left(x_{2}, x_{1}\right) & =d y\left(f \left(x_{2}\right), f\left(x_{1}\right)\right) \\ & =d y \left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \end{aligned}`	$\begin{aligned} d\left(x_{2}, x_{1}\right) & =d y\left(f \left(x_{2}\right), f\left(x_{1}\right)\right) \\ & =d y \left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \end{aligned}$	$\begin{aligned} d(x_{2}, x_{1}) &= d_{\mathcal{Y}}(f(x_{2}), f(x_{1})) \\ &= d_{\mathcal{Y}}(f(x_{1}), f(x_{2})) \\ &= d_{\mathcal{X}}(x_{1}, x_{2}) \end{aligned}$
rph8-g15q_EQ0061 (B12)	019	■ 0.890 ● W|+0	`\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}`	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}$	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & X \otimes Z, & Z \otimes X, \\ Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, & X \otimes Y, & Y \otimes X, \end{array}$
rph8-g15q_EQ0019 (19)	005	■ 0.921 ● W|+0	`\begin{array}{lclc} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}`	$\begin{array}{lclc} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}$	$\begin{array}{llll} Z \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Z, & Y \otimes \mathbb{1}_{2}, & \mathbb{1}_{2} \otimes Y, \\ X \otimes Z, & Z \otimes X, & X \otimes Y, & Y \otimes X, \end{array}$
rph8-g15q_EQ0049 (A16)	018	■ 0.933 ● N|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i\mathcal{L}(x)},$
rph8-g15q_EQ0007 (7)	005	■ 0.945 ● C|+51	`\begin{array}{ll} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], & \alpha_{(1,0)}= \left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right], \\ \alpha_{(0,1)}=\left[\begin{array}{rr} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}= \left[\begin{array}{rr} 0 & -1 \\ -1 & 0 \end{array} \right] . \end{array}`	*(not rendered)*	$\begin{array}{ll} \alpha_{(0,0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, & \alpha_{(1,0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \\ \alpha_{(0,1)} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, & \alpha_{(1,1)} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}. \end{array}$
rph8-g15q_EQ0064 (B9)	019	■ 0.951 ● N|+1	`\begin{aligned} d_{A}\left(x_{1}, x_{2}\right) & =\sqrt{ \left(x_{1}-x_{2}\right)^{T} V^{T} B V\left(x_{1}-x_{2} \right)} \\ & =\sqrt{\left(V x_{1}-V x_{2}\right)^{T} B \left(V x_{1}-V x_{2}\right)} \\ & =d_{B}\left(V x_{1}, V x_{2}\right) \end{aligned}`	$\begin{aligned} d_{A}\left(x_{1}, x_{2}\right) & =\sqrt{ \left(x_{1}-x_{2}\right)^{T} V^{T} B V\left(x_{1}-x_{2} \right)} \\ & =\sqrt{\left(V x_{1}-V x_{2}\right)^{T} B \left(V x_{1}-V x_{2}\right)} \\ & =d_{B}\left(V x_{1}, V x_{2}\right) \end{aligned}$	$\begin{aligned} d_{A}(x_{1}, x_{2}) &= \sqrt{(x_{1}-x_{2})^{T} V^{T} B V (x_{1}-x_{2})} \\ &= \sqrt{(V x_{1}-V x_{2})^{T} B (V x_{1}-V x_{2})} \\ &= d_{B}(V x_{1}, V x_{2}), \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>rph8-g15q_EQ0008 (8)</td><td>005</td><td>0.962
W|+2</td><td>\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\operatorname{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\operatorname{SWAP}(X \otimes X), \end{array}</td><td>$\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\operatorname{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\operatorname{SWAP}(X \otimes X), \end{array}$</td><td>$\begin{array}{ll} V_{(0,0)}=\mathbb{1}_{2} \otimes \mathbb{1}_{2}, & V_{(1,0)}=\mathtt{SWAP}, \\ V_{(0,1)}=X \otimes X, & V_{(1,1)}=\mathtt{SWAP}(X \otimes X), \end{array}$</td></tr>
<tr><td>rph8-g15q_EQ0046 (A13)</td><td>017</td><td>0.985
W|+0</td><td>\begin{aligned} y & =\left.\frac{d}{d t} \exp (t y) \right|_{t=0}=\left.\frac{d}{d t}(f \circ \gamma) \right|_{t=0} \quad \text { by Eq. (A12) } \\ & = \left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \gamma \right|_{t=0}\right) \quad \text { by the chain rule } \\ & =\left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \exp (t x)\right|_{t=0}\right) \quad \text { by Eq. (A11) } \\ & =\mathcal{M}(x) \quad \text { by definition of } \mathcal{M} \end{aligned}</td><td>$\begin{aligned} y & =\left.\frac{d}{d t} \exp (t y) \right|_{t=0}=\left.\frac{d}{d t}(f \circ \gamma) \right|_{t=0} \quad \text { by Eq. (A12) } \\ & = \left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \gamma \right|_{t=0}\right) \quad \text { by the chain rule } \\ & =\left.D f\right|_{1_{G}}\left(\left.\frac{d}{d t} \exp (t x)\right|_{t=0}\right) \quad \text { by Eq. (A11) } \\ & =\mathcal{M}(x) \quad \text { by definition of } \mathcal{M} \end{aligned}$</td><td>$\begin{aligned} y & =\frac{d}{d t} \exp (t y)|_{t=0}=\frac{d}{d t}(f \circ \gamma)|_{t=0} \quad \text { by Eq. (A12) } \\ & = D f|_{1_{G}}\left(\frac{d}{d t} \gamma|_{t=0}\right) \quad \text { by the chain rule } \\ & =D f|_{1_{G}}\left(\frac{d}{d t} \exp (t x)|_{t=0}\right) \quad \text { by Eq. (A11) } \\ & =\mathcal{M}(x) \quad \text { by definition of } \mathcal{M}. \end{aligned}$ (A</td></tr>
<tr><td>rph8-g15q_EQ0067 (D3)</td><td>019</td><td>0.991
N|+1</td><td>\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & \geqslant d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{3}\right)\right)+d_{\mathcal{Y}}\left(f \left(x_{3}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{\mathcal{X}} \left(x_{3}, x_{2}\right) \end{aligned}</td><td>$\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & \geqslant d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{3}\right)\right)+d_{\mathcal{Y}}\left(f \left(x_{3}\right), f\left(x_{2}\right)\right) \\ & =d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{\mathcal{X}} \left(x_{3}, x_{2}\right) \end{aligned}$</td><td>$\begin{aligned} d_{\mathcal{X}}(x_{1}, x_{2}) & \geqslant d_{\mathcal{Y}}(f(x_{1}), f(x_{3}))+d_{\mathcal{Y}}(f(x_{3}), f(x_{2})) \\ & =d_{\mathcal{X}}(x_{1}, x_{3})+d_{\mathcal{X}}(x_{3}, x_{2}), \end{aligned}$</td></tr>
<tr><td>rph8-g15q_EQ0027 (27)</td><td>010</td><td>0.994
K|+0</td><td>d_{\mathrm{Tr}}(\rho, \sigma)=\|\rho-\sigma\|_{1} \equiv \operatorname{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .</td><td>$d_{\mathrm{Tr}}(\rho, \sigma)=\|\rho-\sigma\|_{1} \equiv \operatorname{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .$</td><td>$d_{\mathrm{Tr}}(\rho, \sigma)=\|\rho-\sigma\|_{1} \equiv \operatorname{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right].$</td></tr>
<tr><td>rph8-g15q_EQ0054 (B5)</td><td>018</td><td>0.994
K|+0</td><td>\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle</td><td>$\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle$</td><td>$|\psi_{0}\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle$</td></tr>
<tr><td>rph8-g15q_EQ0009 (9)</td><td>005</td><td>0.996
K|+0</td><td>U\left(x_{1}, x_{2}\right)=R_{Z}\left(x_{1}\right) \otimes R_{Z}\left(x_{2}\right),</td><td>$U\left(x_{1}, x_{2}\right)=R_{Z}\left(x_{1}\right) \otimes R_{Z}\left(x_{2}\right),$</td><td>$U(x_{1}, x_{2})=R_{Z}(x_{1}) \otimes R_{Z}(x_{2}),$</td></tr>
<tr><td>rph8-g15q_EQ0011 (11)</td><td>005</td><td>0.999
K|+0</td><td>\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle,</td><td>$\left|\psi_{0}\right\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle,$</td><td>$|\psi_{0}\rangle=\sqrt{p}|+,+\rangle- \sqrt{1-p}|-,-\rangle,$</td></tr>
<tr><td>rph8-g15q_EQ0005 (5)</td><td>005</td><td>1.000
K|+0</td><td>y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right),</td><td>$y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right),$</td><td>$y(x_{1}, x_{2})=y(x_{2}, x_{1})=y(-x_{1},-x_{2}),$</td></tr>
<tr><td>rph8-g15q_EQ0016 (16)</td><td>005</td><td>1.000
N|+0</td><td>V_{g} e^{-i \mathcal{L}(x)} V_{g}^{\dagger}=e^{-i \mathcal{L}\left(\alpha_{g}(x)\right)}</td><td>$V_{g} e^{-i \mathcal{L}(x)} V_{g}^{\dagger}=e^{-i \mathcal{L}\left(\alpha_{g}(x)\right)}$</td><td>$V_{g} e^{-i \mathcal{L}(x)} V_{g}^{\dagger}=e^{-i \mathcal{L}(\alpha_{g}(x))}$</td></tr>
<tr><td>rph8-g15q_EQ0023 (23)</td><td>007</td><td>1.000
N|+0</td><td>x \mapsto \chi(x)=\sum_{n=0}^{2^{d}-1} x_{n}|n\rangle \mapsto|\Phi(x)\rangle=\frac{\chi(x)}{\|\chi(x)\|},</td><td>$x \mapsto \chi(x)=\sum_{n=0}^{2^{d}-1} x_{n}|n\rangle \mapsto|\Phi(x)\rangle=\frac{\chi(x)}{\|\chi(x)\|},$</td><td>$x \mapsto \chi(x)=\sum_{n=0}^{2^{d}-1} x_{n}|n\rangle \mapsto|\Phi(x)\rangle=\frac{\chi(x)}{\|\chi(x)\|},$</td></tr>
<tr><td>rph8-g15q_EQ0029 (29)</td><td>010</td><td>1.000
K|+0</td><td>d_{B}(\rho, \sigma)=\sqrt{2(1-\sqrt{F(\rho, \sigma)})},</td><td>$d_{B}(\rho, \sigma)=\sqrt{2(1-\sqrt{F(\rho, \sigma)})},$</td><td>$d_{B}(\rho, \sigma)=\sqrt{2\big(1-\sqrt{F(\rho, \sigma)}\big)},$</td></tr>
<tr><td>rph8-g15q_EQ0030 (30)</td><td>010</td><td>1.000
N|+1</td><td>F(\rho, \sigma)=(\operatorname{Tr}[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}])^{2}</td><td>$F(\rho, \sigma)=(\operatorname{Tr}[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}])^{2}$</td><td>$F(\rho, \sigma)=\left(\operatorname{Tr}\left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}\right]\right)^{2}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>rph8-g15q_
EQ0032 (32)</td><td>013</td><td>1.000
K|+0</td><td><code>F\left(\operatorname{id}_{(\mathcal{X}, \alpha)}
\right)=\operatorname{id}_{(\mathcal{X}, \alpha)}=
\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F(
\mathcal{X}, \alpha)} .</code></td><td>$F\left(\operatorname{id}_{(\mathcal{X}, \alpha)}\right)=\operatorname{id}_{(\mathcal{X}, \alpha)}=\operatorname{id}_{\mathcal{X}}=\operatorname{id}_{F(\mathcal{X}, \alpha)} .$</td><td>$F(\mathrm{id}_{(\mathcal{X},\alpha)}) = \mathrm{id}_{(\mathcal{X},\alpha)} = \mathrm{id}_{\mathcal{X}} = \mathrm{id}_{F(\mathcal{X},\alpha)}.$</td></tr>
<tr><td>rph8-g15q_EQ0001 (1)</td><td>003</td><td>1.000
K|+0</td><td><code>\rho(x)=U(x)\left|\psi_{0}\right\rangle\left\langle
\psi_{0}\right| U(x)^{\dagger} .</code></td><td>$\rho(x)=U(x)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U(x)^{\dagger} .$</td><td>$\rho(x) = U(x)|\psi_0\rangle\langle\psi_0|U(x)^{\dagger}.$</td></tr>
<tr><td>rph8-g15q_EQ0002 (2)</td><td>004</td><td>1.000
N|+0</td><td><code>\operatorname{Ad}_{V_{g}}(\sigma):=V_{g} \sigma V_{g}^{
\dagger}</code></td><td>$\operatorname{Ad}_{V_{g}}(\sigma):=V_{g} \sigma V_{g}^{\dagger}$</td><td>$\mathrm{Ad}_{V_g}(\sigma) := V_g\sigma V_g^{\dagger}$</td></tr>
<tr><td>rph8-g15q_EQ0003 (3)</td><td>004</td><td>1.000
N|+0</td><td><code>\rho\left(\alpha_{g}(x)\right)=V_{g} \rho(x) V_{g}^{
\dagger}</code></td><td>$\rho\left(\alpha_{g}(x)\right)=V_{g} \rho(x) V_{g}^{\dagger}$</td><td>$\rho(\alpha_g(x)) = V_g\rho(x)V_g^{\dagger}$</td></tr>
<tr><td>rph8-g15q_EQ0004 (4)</td><td>004</td><td>1.000
N|+0</td><td><code>f\left(\alpha_{g}(x)\right)=\beta_{g}(f(x))</code></td><td>$f\left(\alpha_{g}(x)\right)=\beta_{g}(f(x))$</td><td>$f(\alpha_g(x)) = \beta_g(f(x))$</td></tr>
<tr><td>rph8-g15q_EQ0006 (6)</td><td>005</td><td>1.000
N|+0</td><td><code>G=\mathbb{Z}_{2} \times \mathbb{Z}_{2}=
\{(0,0),(1,0),(0,1),(1,1)\}</code></td><td>$G=\mathbb{Z}_{2} \times \mathbb{Z}_{2}=\{(0,0),(1,0),(0,1),(1,1)\}$</td><td>$G = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0),(1,0),(0,1),(1,1)\}$</td></tr>
<tr><td>rph8-g15q_
EQ0010 (10)</td><td>005</td><td>1.000
N|+1</td><td><code>R_{Z}(\theta)=e^{-i \theta Z / 2}=\left[\begin{array}{cc}
e^{-i \theta / 2} & 0 \\ 0 & e^{i \theta / 2} \end{array}
\right]</code></td><td>$R_{Z}(\theta)=e^{-i \theta Z / 2}=\left[\begin{array}{cc} e^{-i \theta / 2} & 0 \\ 0 & e^{i \theta / 2} \end{array}\right]$</td><td>$R_Z(\theta) = e^{-i\theta Z/2} = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix},$</td></tr>
<tr><td>rph8-g15q_
EQ0012 (12)</td><td>005</td><td>1.000
N|-1</td><td><code>|+\rangle=\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)
\quad \text { and } \quad|-\rangle=\frac{1}{\sqrt{2}}(|0
\rangle-|1\rangle)</code></td><td>$|+\rangle=\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle) \quad \text { and } \quad|-\rangle=\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)$</td><td>$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$</td></tr>
<tr><td>rph8-g15q_
EQ0013 (13)</td><td>005</td><td>1.000
K|+0</td><td><code>\rho(x)=U\left(x_{1}, x_{2}\right)\left|\psi_{0}\right
\rangle\left\langle\psi_{0}\right| U\left(x_{1}, x_{2}
\right)^{\dagger} .</code></td><td>$\rho(x)=U\left(x_{1}, x_{2}\right)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U\left(x_{1}, x_{2}\right)^{\dagger} .$</td><td>$\rho(x) = U(x_1,x_2)|\psi_0\rangle\langle\psi_0|U(x_1,x_2)^{\dagger}.$</td></tr>
<tr><td>rph8-g15q_
EQ0014 (14)</td><td>005</td><td>1.000
N|+0</td><td><code>U(x)=e^{-i \mathcal{L}(x)},</code></td><td>$U(x)=e^{-i \mathcal{L}(x)},$</td><td>$U(x) = e^{-i\mathcal{L}(x)},$</td></tr>
<tr><td>rph8-g15q_
EQ0015 (15)</td><td>005</td><td>1.000
N|-1</td><td><code>L_{1}:=\mathcal{L}\left(e_{1}\right) \quad \text { and }
\quad L_{2}:=\mathcal{L}\left(e_{2}\right),</code></td><td>$L_{1}:=\mathcal{L}\left(e_{1}\right) \quad \text { and } \quad L_{2}:=\mathcal{L}\left(e_{2}\right),$</td><td>$L_1 := \mathcal{L}(e_1) \quad \text{and} \quad L_2 := \mathcal{L}(e_2),$</td></tr>
<tr><td>rph8-g15q_
EQ0017 (17)</td><td>005</td><td>1.000
N|+0</td><td><code>V_{g} \mathcal{L}(x) V_{g}^{\dagger}=\mathcal{L}\left(
\alpha_{g}(x)\right)</code></td><td>$V_{g} \mathcal{L}(x) V_{g}^{\dagger}=\mathcal{L}\left(\alpha_{g}(x)\right)$</td><td>$V_g\mathcal{L}(x)V_g^{\dagger} = \mathcal{L}(\alpha_g(x))$</td></tr>
<tr><td>rph8-g15q_
EQ0020 (20)</td><td>006</td><td>1.000
N|-1</td><td><code>L_{1}=\frac{Z}{2} \otimes \mathbb{1}_{2} \quad \text {
and } \quad L_{2}=\mathbb{1}_{2} \otimes \frac{Z}{2}</code></td><td>$L_{1}=\frac{Z}{2} \otimes \nVdash_{2} \quad \text { and } \quad L_{2}=\nVdash_{2} \otimes \frac{Z}{2}$</td><td>$L_1 = \frac{Z}{2} \otimes \mathbb{1}_2 \quad \text{and} \quad L_2 = \mathbb{1}_2 \otimes \frac{Z}{2}$</td></tr>
<tr><td>rph8-g15q_
EQ0021 (21)</td><td>006</td><td>1.000
N|+0</td><td><code>x \mapsto|\Phi(x)\rangle:=\left(\bigotimes_{k=1}^{d} \exp
\left(-\frac{i x_{k}}{2} X_{k}\right)\right)|0 \cdots 0
\rangle,</code></td><td>$x \mapsto|\Phi(x)\rangle:=\left(\bigotimes_{k=1}^{d} \exp \left(-\frac{i x_{k}}{2} X_{k}\right)\right)|0 \cdots 0\rangle,$</td><td>$x \mapsto |\Phi(x)\rangle := \left(\bigotimes_{k=1}^{d} \exp\left(-\frac{ix_k}{2}X_k\right)\right)|0\cdots 0\rangle,$</td></tr>
<tr><td>rph8-g15q_
EQ0022 (22)</td><td>006</td><td>1.000
K|+0</td><td><code>\rho(x):=|\Phi(x)\rangle\langle\Phi(x)| .</code></td><td>$\rho(x):=|\Phi(x)\rangle\langle\Phi(x)| .$</td><td>$\rho(x) := |\Phi(x)\rangle\langle\Phi(x)|.$</td></tr>
<tr><td>rph8-g15q_
EQ0024 (24)</td><td>007</td><td>1.000
N|+0</td><td><code>\left(x_{0}, x_{1}, \ldots, x_{N-1}\right) \mapsto \frac{
\left(x_{0}, x_{1}, \ldots, x_{N-1}, 1\right)}{\sqrt{1+
\|x\|^{2}}},</code></td><td>$\left(x_{0}, x_{1}, \ldots, x_{N-1}\right) \mapsto \frac{\left(x_{0}, x_{1}, \ldots, x_{N-1}, 1\right)}{\sqrt{1+\|x\|^{2}}},$</td><td>$(x_0,x_1,\ldots,x_{N-1}) \mapsto \frac{(x_0,x_1,\ldots,x_{N-1},1)}{\sqrt{1+\|x\|^2}},$</td></tr>
<tr><td>rph8-g15q_
EQ0025 (25)</td><td>007</td><td>1.000
C|-4</td><td><code>\left(x_{0}, \ldots, x_{N-1}\right) \mapsto \frac{
\left(e^{x_{0}}, \ldots, e^{x_{N-1}}, 1\right)}{1+
\sum_{j=0}^{N-1} e^{x_{j}}},</code></td><td>$\left(x_{0}, \ldots, x_{N-1}\right) \mapsto \frac{\left(e^{x_{0}}, \ldots, e^{x_{N-1}}, 1\right)}{1+\sum_{j=0}^{N-1} e^{x_{j}}},$</td><td>$(x_0,\ldots,x_{N-1}) \mapsto \frac{(e^{x_0},\ldots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1} e^{x_j}},$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0026 (26)	009	■ 1.000 ● N|+0	`d_{\mathcal{X}}\left(x_{1}, x_{2}\right):=d_{\mathcal{Y}} \left(f\left(x_{1}\right), f\left(x_{2}\right)\right)`	$d_{\mathcal{X}}\left(x_{1}, x_{2}\right):=d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right)$	$d_{\mathcal{X}}(x_1, x_2) := d_{\mathcal{Y}}(f(x_1), f(x_2))$
rph8-g15q_ EQ0028 (28)	010	■ 1.000 ● K|+0	`d_{\mathrm{HS}}(\rho, \sigma)=\sqrt{\operatorname{Tr} \left[(\rho-\sigma)^{\dagger}(\rho-\sigma)\right]} .`	$d_{\mathrm{HS}}(\rho, \sigma)=\sqrt{\operatorname{Tr}\left[(\rho-\sigma)^{\dagger}(\rho-\sigma)\right]}.$	$d_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\mathrm{Tr}\,[(\rho-\sigma)^{\dagger}(\rho-\sigma)]}.$
rph8-g15q_ EQ0031 (31)	013	■ 1.000 ● K|+0	`F(g \circ f)=g \circ f=F(g) \circ F(f) .`	$F(g \circ f)=g \circ f=F(g) \circ F(f).$	$F(g \circ f) = g \circ f = F(g) \circ F(f).$
rph8-g15q_ EQ0033 (33)	014	■ 1.000 ● N|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i\mathcal{L}(x)},$
rph8-g15q_ EQ0034 (A1)	016	■ 1.000 ● N|+0	`U(x)=e^{-i \mathcal{L}(x)},`	$U(x)=e^{-i \mathcal{L}(x)},$	$U(x) = e^{-i\mathcal{L}(x)},$
rph8-g15q_ EQ0035 (A2)	016	■ 1.000 ● N|+0	`\gamma(t)=\exp (t M)`	$\gamma(t)=\exp (t M)$	$\gamma(t) = \exp(tM)$
rph8-g15q_ EQ0036 (A3)	016	■ 1.000 ● N|+0	`f(\exp (W))=\exp (\mathcal{M}(W))`	$f(\exp (W))=\exp (\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$
rph8-g15q_ EQ0037 (A4)	016	■ 1.000 ● K|+0	`\left[\mathcal{M}\left(W_{1}\right), \mathcal{M} \left(W_{2}\right)\right]=\mathcal{M}\left(\left[W_{1}, W_{2}\right]\right)`	$\left[\mathcal{M}\left(W_{1}\right), \mathcal{M}\left(W_{2}\right)\right]=\mathcal{M}\left(\left[W_{1}, W_{2}\right]\right)$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$
rph8-g15q_ EQ0038 (A5)	017	■ 1.000 ● N|+0	`U((s, t))=e^{-i s X-i t Y}`	$U((s, t))=e^{-i s X-i t Y}$	$U((s,t)) = e^{-isX-itY}$
rph8-g15q_ EQ0039 (A6)	017	■ 1.000 ● N|+0	`\begin{aligned} U((s, 0)) U((0, t)) & =e^{-i s X} e^{-i t Y} \\ & \neq e^{-i(s X+t Y)} \\ & =U((s, 0)+(0, t)) \end{aligned}`	$\begin{aligned} U((s, 0)) U((0, t)) & =e^{-i s X} e^{-i t Y} \\ & \neq e^{-i(s X+t Y)} \\ & =U((s, 0)+(0, t)) \end{aligned}$	$\begin{aligned} U((s,0))U((0,t)) &= e^{-isX}e^{-itY} \\ &\neq e^{-i(sX+tY)} \\ &= U((s,0)+(0,t)) \end{aligned}$
rph8-g15q_ EQ0040 (A7)	017	■ 1.000 ● N|+0	`\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^{2} \backslash\{0\}, r \in \mathbb{R}\right\},`	$\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^{2} \backslash\{0\}, r \in \mathbb{R}\right\},$	$\mathbb{R}_{\alpha} = \{r\alpha \ : \ \alpha \in \mathbb{R}^2 \setminus \{0\},\ r \in \mathbb{R}\},$
rph8-g15q_ EQ0041 (A8)	017	■ 1.000 ● N|-1	`\begin{aligned} U(r \alpha) U(u \alpha) & =e^{-i r W} e^{-i u W} \\ & =e^{-i(r+u) W} \\ & =U(r \alpha+u \alpha), \end{aligned}`	$\begin{aligned} U(r \alpha) U(u \alpha) & =e^{-i r W} e^{-i u W} \\ & =e^{-i(r+u) W} \\ & =U(r \alpha+u \alpha), \end{aligned}$	$\begin{aligned} U(r\alpha)U(u\alpha) &= e^{-irW}e^{-iuW} \\ &= e^{-i(r+u)W} \\ &= U(r\alpha+u\alpha), \end{aligned}$
rph8-g15q_ EQ0042 (A9)	017	■ 1.000 ● N|+0	`\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{2}\right)`	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{2}\right)$	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^2 \xrightarrow{U} \mathscr{U}(\mathbb{C}^2)$
rph8-g15q_ EQ0043 (A10)	017	■ 1.000 ● N|+0	`f(\exp (x))=\exp (\mathcal{M}(x))`	$f(\exp (x))=\exp (\mathcal{M}(x))$	$f\big(\exp(x)\big) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0044 (A11)	017	■ 1.000 ● N|+0	`\gamma(t)=\exp (t x)`	$\gamma(t)=\exp (t x)$	$\gamma(t) = \exp(tx)$
rph8-g15q_ EQ0045 (A12)	017	■ 1.000 ● K|+0	`(f \circ \gamma)(t)=\exp (t y)`	$(f \circ \gamma)(t)=\exp (t y)$	$(f \circ \gamma)(t) = \exp(ty)$
rph8-g15q_ EQ0047 (A14)	017	■ 1.000 ● K|+0	`f(\exp (t x))=\exp (t \mathcal{M}(x))`	$f(\exp (t x))=\exp (t \mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$
rph8-g15q_ EQ0048 (A15)	018	■ 1.000 ● N|+0	`f(x)=\exp (\mathcal{M}(x))`	$f(x)=\exp (\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0050 (B1)	018	■ 1.000 ● K|+0	`c\left(x_{1}, x_{2}\right)=c\left(x_{2}, x_{1}\right)=c \left(-x_{1},-x_{2}\right) .`	$c\left(x_{1}, x_{2}\right)=c\left(x_{2}, x_{1}\right)=c\left(-x_{1},-x_{2}\right).$	$c(x_1, x_2) = c(x_2, x_1) = c(-x_1, -x_2).$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
rph8-g15q_ EQ0051 (B2)	018	■ 1.000 ● W|+0	`c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}`	$c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$	$c(x)= \begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$
rph8-g15q_ EQ0052 (B3)	018	■ 1.000 ● K|+0	`y(x)=\operatorname{Tr}[\rho(x) O],`	$y(x)=\operatorname{Tr}[\rho(x) O],$	$y(x)=\operatorname{Tr}[\rho(x) O],$
rph8-g15q_ EQ0053 (B4)	018	■ 1.000 ● K|+0	`y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right)`	$y\left(x_{1}, x_{2}\right)=y\left(x_{2}, x_{1}\right)=y \left(-x_{1},-x_{2}\right)$	$y(x_{1}, x_{2})=y(x_{2}, x_{1})=y(-x_{1},-x_{2})$
rph8-g15q_ EQ0055 (B6)	018	■ 1.000 ● K|+0	`O=X \otimes X,`	$O=X \otimes X,$	$O=X \otimes X,$
rph8-g15q_ EQ0056 (B7)	018	■ 1.000 ● K|+0	`y\left(x_{1}, x_{2}\right)=\cos \left(x_{1}\right) \sin \left(x_{2}\right)+2 \sqrt{p(1-p)} \sin \left(x_{1} \right) \sin \left(x_{2}\right) .`	$y\left(x_{1}, x_{2}\right)=\cos \left(x_{1}\right) \sin \left(x_{2}\right)+2 \sqrt{p(1-p)} \sin \left(x_{1} \right) \sin \left(x_{2}\right) .$	$y(x_{1}, x_{2})=\cos(x_{1}) \sin(x_{2})+2 \sqrt{p(1-p)} \sin(x_{1}) \sin(x_{2}).$
rph8-g15q_ EQ0057 (B8)	018	■ 1.000 ● N|-1	`L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],`	$L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$	$L_{1}=\left[\begin{array}{llll} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$
rph8-g15q_ EQ0058 (C3)	019	■ 1.000 ● W|+0	`\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .`	$\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .$	$\left[\begin{array}{llll} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right]=-\left[\begin{array}{llll} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right] .$
rph8-g15q_ EQ0059 (B10)	019	■ 1.000 ● N|+0	`L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],`	$L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$	$L_{1}=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$
rph8-g15q_ EQ0060 (B11)	019	■ 1.000 ● N|+0	`a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .`	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .$	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i \mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R} .$
rph8-g15q_ EQ0062 (C1)	019	■ 1.000 ● K|+0	`d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=\sqrt{ \left(x_{1}-x_{2}\right)^{T}\left(V^{T} V\right) \left(x_{1}-x_{2}\right)},`	$d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=\sqrt{ \left(x_{1}-x_{2}\right)^{T}\left(V^{T} V\right) \left(x_{1}-x_{2}\right)},$	$d_{\mathcal{X}}(x_{1}, x_{2})=\sqrt{(x_{1}-x_{2})^{T}(V^{T} V)(x_{1}-x_{2})},$
rph8-g15q_ EQ0063 (C2)	019	■ 1.000 ● K|+0	`d_{A}\left(x_{1}, x_{2}\right):=\sqrt{\left(x_{1}-x_{2} \right)^{T} A\left(x_{1}-x_{2}\right)} .`	$d_{A}\left(x_{1}, x_{2}\right):=\sqrt{\left(x_{1}-x_{2} \right)^{T} A\left(x_{1}-x_{2}\right)} .$	$d_{A}(x_{1}, x_{2}):=\sqrt{(x_{1}-x_{2})^{T} A(x_{1}-x_{2})}.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

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rph8-g15q_ EQ0066 (D2)	019	■ 1.000 ● W|-1	`\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & =d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & \geqslant d_{\mathcal{Y}}\left(f \left(x_{1}\right), y\right)+d_{\mathcal{Y}}\left(y, f \left(x_{2}\right)\right) \end{aligned}`	$\begin{aligned} d_{\mathcal{X}}\left(x_{1}, x_{2}\right) & =d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \\ & \geqslant d_{\mathcal{Y}}\left(f \left(x_{1}\right), y\right)+d_{\mathcal{Y}}\left(y, f \left(x_{2}\right)\right) \end{aligned}$	$\begin{aligned} d_{\mathcal{X}}(x_1, x_2) & = d_{\mathcal{Y}}(f(x_1), f(x_2)) \\ & \geqslant d_{\mathcal{Y}}(f(x_1), y) + d_{\mathcal{Y}}(y, f(x_2)) \end{aligned}$
rph8-g15q_ EQ0068 (D4)	019	■ 1.000 ● N|+0	`0=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right),`	$0=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_{\mathcal{Y}}(f(x_1), f(x_2)),$
rph8-g15q_ EQ0069 (D5)	020	■ 1.000 ● N|+0	`c(x)= \begin{cases}+1, & \text { if } x \in A, \\ -1, & \text { if } x \in B .\end{cases}`	$c(x)= \begin{cases}+1, & \text { if } x \in A, \\ -1, & \text { if } x \in B .\end{cases}$	$c(x) = \begin{cases}+1, & \text{if } x \in A, \\ -1, & \text{if } x \in B.\end{cases}$
rph8-g15q_ EQ0070 (D6)	020	■ 1.000 ● N|-1	`\rho_{A}:=\frac{1}{\# A} \sum_{a \in A} \rho(a) \quad \text { and } \quad \rho_{B}:=\frac{1}{\# B} \sum_{b \in B} \rho(b),`	$\rho_{A}:=\frac{1}{\# A} \sum_{a \in A} \rho(a) \quad \text { and } \quad \rho_{B}:=\frac{1}{\# B} \sum_{b \in B} \rho(b),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$
rph8-g15q_ EQ0071 (D7)	020	■ 1.000 ● K|+0	`C\left(\rho_{A}, \rho_{B}\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_{A}, \rho_{B}\right) .`	$C\left(\rho_{A}, \rho_{B}\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_{A}, \rho_{B}\right) .$	$C(\rho_A, \rho_B) = 1 - \frac{1}{2} d_{\mathrm{HS}}(\rho_A, \rho_B).$
rph8-g15q_ EQ0072 (D8)	020	■ 1.000 ● K|+0	`U_{\theta}(x)=\left(\prod_{n=1}^{4} R_{X}(x) R_{Y}\left(\theta_{n}\right)\right) R_{X}(x),`	$U_{\theta}(x)=\left(\prod_{n=1}^{4} R_{X}(x) R_{Y}\left(\theta_{n}\right)\right) R_{X}(x),$	$U_\theta(x) = \left(\prod_{n=1}^{4} R_X(x) R_Y(\theta_n)\right) R_X(x),$
rph8-g15q_ EQ0073 (D9)	020	■ 1.000 ● N|+0	`\prod_{n=1}^{N} A_{n}=A_{N} \cdots A_{1} .`	$\prod_{n=1}^{N} A_{n}=A_{N} \cdots A_{1} .$	$\prod_{n=1}^{N} A_n = A_N \cdots A_1.$
rph8-g15q_ EQ0074 (D10)	020	■ 1.000 ● K|+0	`\rho_{\theta}(x)=|x\rangle\langle x|,`	$\rho_{\theta}(x)=\lvert x\rangle\langle x\rvert,$	$\rho_\theta(x) = \lvert x\rangle\langle x\rvert,$
rph8-g15q_ EQ0075 (D11)	020	■ 1.000 ● K|+0	`|x\rangle=U_{\theta}(x)|0\rangle .`	$\lvert x\rangle=U_{\theta}(x)\lvert 0\rangle .$	$\lvert x\rangle = U_\theta(x)\lvert 0\rangle.$
rph8-g15q_ EQ0076 (D12)	020	■ 1.000 ● K|+0	`O:=\rho_{A}-\rho_{B},`	$O:=\rho_{A}-\rho_{B},$	$O := \rho_A - \rho_B,$
rph8-g15q_ EQ0077 (D13)	020	■ 1.000 ● K|+0	`y(x)=\operatorname{Tr}[\rho(x) O] .`	$y(x)=\operatorname{Tr}[\rho(x) O] .$	$y(x) = \mathrm{Tr}\big[\rho(x) O\big].$
rph8-g15q_ EQ0078 (D14)	020	■ 1.000 ● K|+0	`\operatorname{Tr}\left[\rho_{A}-\rho_{B}\right]=\operatorname{Tr}\left[\rho_{A}\right]-\operatorname{Tr}\left[\rho_{B}\right]=1-1=0,`	$\operatorname{Tr}\left[\rho_{A}-\rho_{B}\right]=\operatorname{Tr}\left[\rho_{A}\right]-\operatorname{Tr}\left[\rho_{B}\right]=1-1=0,$	$\mathrm{Tr}[\rho_A - \rho_B] = \mathrm{Tr}[\rho_A] - \mathrm{Tr}[\rho_B] = 1 - 1 = 0,$
rph8-g15q_ EQ0079 (D15)	020	■ 1.000 ● N|+0	`\operatorname{sgn}(y(x))=c(x)`	$\operatorname{sgn}(y(x))=c(x)$	$\mathrm{sgn}\big(y(x)\big) = c(x)$
rph8-g15q_ EQ0080 (E2)	021	■ 1.000 ● W|+1	`\overbrace{\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}}^{n \text { times }} \cong \mathbb{C}^{2^{n}},`	$\overbrace{\mathbb{C}^{2} \otimes \cdots \otimes \mathbb{C}^{2}}^{n \text { times }} \cong \mathbb{C}^{2^{n}},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text{ times}} \cong \mathbb{C}^{2^n},$
rph8-g15q_ EQ0081 (E3)	021	■ 1.000 ● N|+0	`\left|x_{0} x_{1} \cdots x_{n-1}\right\rangle=\left|x_{0}\right\rangle \otimes\left|x_{1}\right\rangle \otimes \cdots \otimes\left|x_{n-1}\right\rangle .`	$\left\lvert x_{0} x_{1} \cdots x_{n-1}\right\rangle=\left\lvert x_{0}\right\rangle \otimes\left\lvert x_{1}\right\rangle \otimes \cdots \otimes\left\lvert x_{n-1}\right\rangle .$	$\lvert x_0 x_1 \cdots x_{n-1}\rangle = \lvert x_0\rangle \otimes \lvert x_1\rangle \otimes \cdots \otimes \lvert x_{n-1}\rangle.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00001	001	—	`{ }^{\text {® }}`	${ }^{\text {® }}$
rph8-g15q_F00002	001	—	`{ }^{1,2, *}`	${ }^{1,2, *}$
rph8-g15q_F00003	001	—	`{ }^{3,4, \dagger}`	${ }^{3,4, \dagger}$
rph8-g15q_F00004	001	—	`{ }^{5, \ddagger}`	${ }^{5, \ddagger}$
rph8-g15q_F00005		—	`{ }^{\dagger}`	${ }^{\dagger}$
rph8-g15q_F00006		—	`{ }^{\ddagger}`	${ }^{\ddagger}$
rph8-g15q_F00007	003	—	`x`	x
rph8-g15q_F00008	003	—	`\theta`	θ
rph8-g15q_F00009	003	—	`\rangle`	$\rangle$
rph8-g15q_F00010	003	—	`v=\left(v_{1}, \ldots, v_{d}\right)`	$v=\left(v_{1}, \ldots, v_{d}\right)$
rph8-g15q_F00011	003	—	`\mathbb{R}^{d}`	$\mathbb{R}^{d}$
rph8-g15q_F00012	003	—	`\mathcal{H}`	$\mathcal{H}$
rph8-g15q_F00013	003	—	`\mathscr{S}(\mathcal{H})`	$\mathscr{S}(\mathcal{H})$
rph8-g15q_F00014	003	—	`\mathscr{U}(\mathcal{H})`	$\mathscr{U}(\mathcal{H})$
rph8-g15q_F00015	003	—	`\mathcal{X}`	$\mathcal{X}$
rph8-g15q_F00016	003	—	`X`	X
rph8-g15q_F00017	003	—	`\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})`	$\rho: \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00018	003	—	`U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})`	$U: \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})$
rph8-g15q_F00019	003	—	`\left|\psi_{0}\right\rangle \in \mathcal{H}`	$\left\lvert\psi_{0}\right\rangle \in \mathcal{H}$
rph8-g15q_F00020	003	—	`\rho: \mathcal{X} \rightarrow`	$\rho: \mathcal{X} \rightarrow$
rph8-g15q_F00021	004	—	`G`	G
rph8-g15q_F00022	004	—	`\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})`	$\alpha: G \rightarrow \operatorname{Aut}(\mathcal{X})$
rph8-g15q_F00023	004	—	`\operatorname{Aut}(\mathcal{X})`	$\operatorname{Aut}(\mathcal{X})$
rph8-g15q_F00024	004	—	`\alpha`	α
rph8-g15q_F00025	004	—	`(\mathcal{X}, \alpha)`	$(\mathcal{X}, \alpha)$
rph8-g15q_F00026	004	—	`g \in G`	$g \in G$
rph8-g15q_F00027	004	—	`\alpha_{g}: \mathcal{X} \rightarrow \mathcal{X}`	$\alpha_{g}: \mathcal{X} \rightarrow \mathcal{X}$
rph8-g15q_F00028	004	—	`x \in \mathcal{X}`	$x \in \mathcal{X}$
rph8-g15q_F00029	004	—	`\alpha_{g}(x)`	$\alpha_{g}(x)$
rph8-g15q_F00030	004	—	`\alpha_{g h}(x)=\alpha_{g}\left(\alpha_{h}(x)\right)`	$\alpha_{g h}(x)=\alpha_{g}\left(\alpha_{h}(x)\right)$
rph8-g15q_F00031	004	—	`g, h \in G`	$g, h \in G$
rph8-g15q_F00032	004	—	`\alpha_{1_{G}}=\mathrm{id}_{\mathcal{X}}`	$\alpha_{1_{G}}=\mathrm{id}_{\mathcal{X}}$
rph8-g15q_F00033	004	—	`1_{G}`	1_{G}
rph8-g15q_F00034	004	—	`V: G \rightarrow`	$V: G \rightarrow$
rph8-g15q_F00035	004	—	`\operatorname{Aut}(\mathcal{H})`	$\operatorname{Aut}(\mathcal{H})$
rph8-g15q_F00036	004	—	`\operatorname{Ad}_{V_{g}} \in \operatorname{Aut}(\mathscr{S}(\mathcal{H}))`	$\operatorname{Ad}_{V_{g}} \in \operatorname{Aut}(\mathscr{S}(\mathcal{H}))$
rph8-g15q_F00037	004	—	`\alpha_{(1,0)}`	$\alpha_{(1,0)}$
rph8-g15q_F00038	004	—	`x_{2}=x_{1}, \alpha_{(0,1)}`	$x_{2}=x_{1}, \alpha_{(0,1)}$
rph8-g15q_F00039	004	—	`\alpha_{(1,1)}`	$\alpha_{(1,1)}$
rph8-g15q_F00040	004	—	`x_{2}=-x_{1}`	$x_{2}=-x_{1}$
rph8-g15q_F00041	004	—	`\sigma \in \mathscr{S}(\mathcal{H})`	$\sigma \in \mathscr{S}(\mathcal{H})$
rph8-g15q_F00042	004	—	`\left(\mathscr{S}(\mathcal{H}), \operatorname{Ad}_{V}\right)`	$\left(\mathscr{S}(\mathcal{H}), \operatorname{Ad}_{V}\right)$
rph8-g15q_F00043	004	—	`V: G \rightarrow \mathscr{U}(\mathcal{H})`	$V: G \rightarrow \mathscr{U}(\mathcal{H})$
rph8-g15q_F00044	004	—	`\rho`	ρ
rph8-g15q_F00045	004	—	`\mathcal{X}, \alpha`	$\mathcal{X}, \alpha$
rph8-g15q_F00046	004	—	`\mathcal{Y}, \beta`	$\mathcal{Y}, \beta$
rph8-g15q_F00047	004	—	`(\mathcal{Y}, \beta)`	$(\mathcal{Y}, \beta)$
rph8-g15q_F00048	004	—	`f: \mathcal{X} \rightarrow`	$f: \mathcal{X} \rightarrow$
rph8-g15q_F00049	004	—	`\mathcal{Y}`	$\mathcal{Y}$
rph8-g15q_F00050	005	—	`\mathcal{X}=\mathbb{R}^{2}`	$\mathcal{X}=\mathbb{R}^{2}$
rph8-g15q_F00051	005	—	`y: \mathcal{X} \rightarrow \mathbb{R}`	$y: \mathcal{X} \rightarrow \mathbb{R}$
rph8-g15q_F00052	005	—	`y(x)>0`	$y(x)>0$
rph8-g15q_F00053	005	—	`y(x)<0`	$y(x)<0$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00054	005	—	`G \rightarrow \operatorname{Aut}\left(\mathbb{R}^{2}\right)`	$G \rightarrow \operatorname{Aut}\left(\mathbb{R}^{2}\right)$
rph8-g15q_F00055	005	—	`\mathcal{H}=\mathbb{C}^{2} \otimes \mathbb{C}^{2}`	$\mathcal{H}=\mathbb{C}^{2} \otimes \mathbb{C}^{2}$
rph8-g15q_F00056	005	—	`X=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right], \mathbb{1}_{2}`	$X=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right], \mathbb{1}_{2}$
rph8-g15q_F00057	005	—	`2 \times 2`	2×2
rph8-g15q_F00058	005	—	`\operatorname{SWAP}(|\psi\rangle \otimes|\phi\rangle)=|\phi\rangle \otimes|\psi\rangle`	$\operatorname{SWAP}(\vert\psi\rangle \otimes\vert\phi\rangle)=\vert\phi\rangle \otimes\vert\psi\rangle$
rph8-g15q_F00059	005	—	`|\psi\rangle,|\phi\rangle \in \mathbb{C}^{2}`	$\vert\psi\rangle,\vert\phi\rangle \in \mathbb{C}^{2}$
rph8-g15q_F00060	005	—	`V_{g} \rho(x) V_{g}^{\dagger}=\rho\left(\alpha_{g}(x)\right)`	$V_{g} \rho(x) V_{g}^{\dagger}=\rho\left(\alpha_{g}(x)\right)$
rph8-g15q_F00061	005	—	`x=\left(x_{1}, x_{2}\right) \in \mathcal{X}`	$x=\left(x_{1}, x_{2}\right) \in \mathcal{X}$
rph8-g15q_F00062	005	—	`Z`	Z
rph8-g15q_F00063	005	—	`Z=\left[\begin{array}{cc}1 & 0 \\ 0 & -1\end{array}\right]`	$Z=\left[\begin{array}{cc}1 & 0 \\ 0 & -1\end{array}\right]$
rph8-g15q_F00064	005	—	`0<p<1`	$0<p<1$
rph8-g15q_F00065	005	—	`p`	p
rph8-g15q_F00066	005	—	`| \pm, \pm\rangle:=| \pm\rangle \otimes| \pm\rangle`	$\vert \pm, \pm\rangle:=\vert \pm\rangle \otimes\vert \pm\rangle$
rph8-g15q_F00067	005	—	`x=\left(x_{1}, x_{2}\right)`	$x=\left(x_{1}, x_{2}\right)$
rph8-g15q_F00068	005	—	`\mathcal{L}: \mathbb{R}^{2} \rightarrow \mathbb{M}_{4}`	$\mathcal{L}: \mathbb{R}^{2} \rightarrow \mathbb{M}_{4}$
rph8-g15q_F00069	005	—	`\mathcal{L}(x) \in \mathbb{M}_{4}`	$\mathcal{L}(x) \in \mathbb{M}_{4}$
rph8-g15q_F00070	005	—	`\mathbb{M}_{4}`	$\mathbb{M}_{4}$
rph8-g15q_F00071	005	—	`-i \mathcal{L}(x) \in \mathfrak{s u}(4)`	$-i \mathcal{L}(x) \in \mathfrak{s u}(4)$
rph8-g15q_F00072	005	—	`4 \times 4`	4×4
rph8-g15q_F00073	005	—	`\mathcal{L}`	$\mathcal{L}$
rph8-g15q_F00074	005	—	`e_{1}=(1,0)`	$e_{1}=(1,0)$
rph8-g15q_F00075	005	—	`e_{2}=(0,1)`	$e_{2}=(0,1)$
rph8-g15q_F00076	005	—	`2\left(4^{2}-1\right)=30`	$2\left(4^{2}-1\right)=30$
rph8-g15q_F00077	005	—	`\mathcal{L}(x)`	$\mathcal{L}(x)$
rph8-g15q_F00078	005	—	`V_{g}`	V_{g}
rph8-g15q_F00079	005	—	`g=(1,0)`	$g=(1,0)$
rph8-g15q_F00080	005	—	`\mathrm{SWAP} L_{1} \mathrm{SWAP}=L_{2}`	$\mathrm{SWAP} L_{1} \mathrm{SWAP}=L_{2}$
rph8-g15q_F00081	005	—	`\mathrm{SWAP} L_{2} \mathrm{SWAP}=L_{1}`	$\mathrm{SWAP} L_{2} \mathrm{SWAP}=L_{1}$
rph8-g15q_F00082	005	—	`g=(0,1)`	$g=(0,1)$
rph8-g15q_F00083	005	—	`(X \otimes X) L_{1}(X \otimes X)=-L_{1}`	$(X \otimes X) L_{1}(X \otimes X)=-L_{1}$
rph8-g15q_F00084	005	—	`(X \otimes X) L_{2}(X \otimes`	$(X \otimes X) L_{2}(X \otimes$
rph8-g15q_F00085	005	—	`X)=-L_{2}`	$X)=-L_{2}$
rph8-g15q_F00086	005	—	`\left(L_{1}, L_{2}\right)`	$\left(L_{1}, L_{2}\right)$
rph8-g15q_F00087	005	—	`L_{1}`	L_{1}
rph8-g15q_F00088	006	—	`\tau_{\mathcal{X}}`	$\tau_{\mathcal{X}}$
rph8-g15q_F00089	006	—	`\left(\mathcal{X}, \tau_{\mathcal{X}}\right)`	$\left(\mathcal{X}, \tau_{\mathcal{X}}\right)$
rph8-g15q_F00090	006	—	`\mathcal{A}`	$\mathcal{A}$
rph8-g15q_F00091	006	—	`\left(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A}\right)`	$\left(\mathcal{X}, \tau_{\mathcal{X}}, \mathcal{A}\right)$
rph8-g15q_F00092	006	—	`\mathcal{H}=\mathbb{C}^{d}`	$\mathcal{H}=\mathbb{C}^{d}$
rph8-g15q_F00093	006	—	`\mathbb{C P}^{d-1}`	$\mathbb{C P}^{d-1}$
rph8-g15q_F00094	006	—	`\mathcal{X}=\mathbb{R}^{d}`	$\mathcal{X}=\mathbb{R}^{d}$
rph8-g15q_F00095	006	—	`x=\left(x_{1}, \ldots, x_{d}\right) \in`	$x=\left(x_{1}, \ldots, x_{d}\right) \in$
rph8-g15q_F00096	006	—	`X_{k}`	X_{k}
rph8-g15q_F00097	006	—	`k`	k
rph8-g15q_F00098	006	—	`\rho: \mathbb{R}^{d} \rightarrow \mathscr{S}\left(\mathbb{C}^{2^{d}}\right)`	$\rho: \mathbb{R}^{d} \rightarrow \mathscr{S}\left(\mathbb{C}^{2^{d}}\right)$
rph8-g15q_F00099	006	—	`Y`	Y
rph8-g15q_F00100	006	—	`\vec{n} \cdot \vec{\sigma}`	$\vec{n} \cdot \vec{\sigma}$
rph8-g15q_F00101	006	—	`\vec{n}`	$\vec{n}$
rph8-g15q_F00102	006	—	`\mathbb{R}^{3}`	$\mathbb{R}^{3}$
rph8-g15q_F00103	006	—	`\vec{\sigma}=(X, Y, Z)`	$\vec{\sigma}=(X, Y, Z)$
rph8-g15q_F00104	006	—	`d`	d
rph8-g15q_F00105	007	—	`x \in \mathbb{R}^{2^{d}} \backslash\{0\}`	$x \in \mathbb{R}^{2^{d}} \backslash\{0\}$
rph8-g15q_F00106	007	—	`\chi(x)`	$\chi(x)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th></tr>
<tr><td>rph8-g15q_F00107</td><td>007</td><td>—</td><td><code>\mathbb{C}^{2^{d}}</code></td><td>$\mathbb{C}^{2^{d}}$</td></tr>
<tr><td>rph8-g15q_F00108</td><td>007</td><td>—</td><td><code>|\Phi(x)\rangle</code></td><td>$|\Phi(x)\rangle$</td></tr>
<tr><td>rph8-g15q_F00109</td><td>007</td><td>—</td><td><code>x=\left(x_{0}, x_{1}, \ldots, x_{2^{d}-1}\right) \in \mathbb{R}^{2^{d}}</code></td><td>$x=\left(x_{0}, x_{1}, \ldots, x_{2^{d}-1}\right) \in \mathbb{R}^{2^{d}}$</td></tr>
<tr><td>rph8-g15q_F00110</td><td>007</td><td>—</td><td><code>n</code></td><td>n</td></tr>
<tr><td>rph8-g15q_F00111</td><td>007</td><td>—</td><td><code>d 0 \mathrm{~s}</code></td><td>$d 0 \mathrm{~s}$</td></tr>
<tr><td>rph8-g15q_F00112</td><td>007</td><td>—</td><td><code>d=2</code></td><td>$d=2$</td></tr>
<tr><td>rph8-g15q_F00113</td><td>007</td><td>—</td><td><code>n \in\{00,01,10,11\}</code></td><td>$n \in\{00,01,10,11\}$</td></tr>
<tr><td>rph8-g15q_F00114</td><td>007</td><td>—</td><td><code>\|\chi(x)\|</code></td><td>$\|\chi(x)\|$</td></tr>
<tr><td>rph8-g15q_F00115</td><td>007</td><td>—</td><td><code>S^{2^{d}-1}</code></td><td>$S^{2^{d}-1}$</td></tr>
<tr><td>rph8-g15q_F00116</td><td>007</td><td>—</td><td><code>\mathbb{R}^{2^{d}}</code></td><td>$\mathbb{R}^{2^{d}}$</td></tr>
<tr><td>rph8-g15q_F00117</td><td>007</td><td>—</td><td><code>\log _{2}\left(2^{d}\right)=d</code></td><td>$\log _{2}\left(2^{d}\right)=d$</td></tr>
<tr><td>rph8-g15q_F00118</td><td>007</td><td>—</td><td><code>S^{2^{d}}</code></td><td>$S^{2^{d}}$</td></tr>
<tr><td>rph8-g15q_F00119</td><td>007</td><td>—</td><td><code>\frac{\pi}{4}</code></td><td>$\frac{\pi}{4}$</td></tr>
<tr><td>rph8-g15q_F00120</td><td>007</td><td>—</td><td><code>\frac{\pi}{2}</code></td><td>$\frac{\pi}{2}$</td></tr>
<tr><td>rph8-g15q_F00121</td><td>007</td><td>—</td><td><code>x_{0}</code></td><td>x_{0}</td></tr>
<tr><td>rph8-g15q_F00122</td><td>007</td><td>—</td><td><code>S^{1}</code></td><td>S^{1}</td></tr>
<tr><td>rph8-g15q_F00123</td><td>007</td><td>—</td><td><code>[-6,6] \cap \mathbb{Z}</code></td><td>$[-6,6] \cap \mathbb{Z}$</td></tr>
<tr><td>rph8-g15q_F00124</td><td>008</td><td>—</td><td><code>\mathbb{R}</code></td><td>$\mathbb{R}$</td></tr>
<tr><td>rph8-g15q_F00125</td><td>008</td><td>—</td><td><code>\left(\mathcal{X}, d_{\mathcal{X}}\right)</code></td><td>$\left(\mathcal{X}, d_{\mathcal{X}}\right)$</td></tr>
<tr><td>rph8-g15q_F00126</td><td>008</td><td>—</td><td><code>d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}</code></td><td>$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$</td></tr>
<tr><td>rph8-g15q_F00127</td><td>010</td><td>—</td><td><code>d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=0</code></td><td>$d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=0$</td></tr>
<tr><td>rph8-g15q_F00128</td><td>010</td><td>—</td><td><code>x_{1}=x_{2}</code></td><td>$x_{1}=x_{2}$</td></tr>
<tr><td>rph8-g15q_F00129</td><td></td><td>—</td><td><code>d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{\mathcal{X}}\left(x_{2}, x_{1}\right)</code></td><td>$d_{\mathcal{X}}\left(x_{1}, x_{2}\right)=d_{\mathcal{X}}\left(x_{2}, x_{1}\right)$</td></tr>
<tr><td>rph8-g15q_F00130</td><td>008</td><td>—</td><td><code>x_{1}, x_{2} \in \mathcal{X}</code></td><td>$x_{1}, x_{2} \in \mathcal{X}$</td></tr>
<tr><td>rph8-g15q_F00131</td><td>008</td><td>—</td><td><code>d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \leqslant d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{ \mathcal{X}}\left(x_{2}, x_{3}\right)</code></td><td>$d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \leqslant d_{\mathcal{X}}\left(x_{1}, x_{3}\right)+d_{\mathcal{X}}\left(x_{2}, x_{3}\right)$</td></tr>
<tr><td>rph8-g15q_F00132</td><td>008</td><td>—</td><td><code>x_{1}, x_{2}, x_{3}</code></td><td>x_{1}, x_{2}, x_{3}</td></tr>
<tr><td>rph8-g15q_F00133</td><td>008</td><td>—</td><td><code>\in \mathcal{X}</code></td><td>$\in \mathcal{X}$</td></tr>
<tr><td>rph8-g15q_F00134</td><td>008</td><td>—</td><td><code>d_{\mathcal{X}}</code></td><td>$d_{\mathcal{X}}$</td></tr>
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<tr><td>rph8-g15q_F00136</td><td>008</td><td>—</td><td><code>\left(X, d_{X}\right)</code></td><td>$\left(X, d_{X}\right)$</td></tr>
<tr><td>rph8-g15q_F00137</td><td>008</td><td>—</td><td><code>K_{\bullet}\left(X, d_{X}\right)</code></td><td>$K_{\bullet}\left(X, d_{X}\right)$</td></tr>
<tr><td>rph8-g15q_F00138</td><td>008</td><td>—</td><td><code>H_{\bullet}\left(X, d_{X}\right)</code></td><td>$H_{\bullet}\left(X, d_{X}\right)$</td></tr>
<tr><td>rph8-g15q_F00139</td><td>008</td><td>—</td><td><code>d_{X}</code></td><td>d_{X}</td></tr>
<tr><td>rph8-g15q_F00140</td><td>008</td><td>—</td><td><code>d_{X}\left(x_{1}, x_{2}\right)=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)</code></td><td>$d_{X}\left(x_{1}, x_{2}\right)=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)$</td></tr>
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<tr><td>rph8-g15q_F00142</td><td>008</td><td>—</td><td><code>X, d_{X}</code></td><td>X, d_{X}</td></tr>
<tr><td>rph8-g15q_F00143</td><td>008</td><td>—</td><td><code>H_{\bullet}\left(X, d_{X}\right)[45,86,87]</code></td><td>$H_{\bullet}\left(X, d_{X}\right)[45,86,87]$</td></tr>
<tr><td>rph8-g15q_F00144</td><td>008</td><td>—</td><td><code>\operatorname{dgm}_{\bullet}\left(X, d_{X}\right)</code></td><td>$\operatorname{dgm}_{\bullet}\left(X, d_{X}\right)$</td></tr>
<tr><td>rph8-g15q_F00145</td><td>008</td><td>—</td><td><code>S</code></td><td>S</td></tr>
<tr><td>rph8-g15q_F00146</td><td>008</td><td>—</td><td><code>\epsilon</code></td><td>ϵ</td></tr>
<tr><td>rph8-g15q_F00147</td><td>008</td><td>—</td><td><code>s \in S</code></td><td>$s \in S$</td></tr>
<tr><td>rph8-g15q_F00148</td><td>008</td><td>—</td><td><code>k+1</code></td><td>$k+1$</td></tr>
<tr><td>rph8-g15q_F00149</td><td>008</td><td>—</td><td><code>v_{1}, v_{2}, \ldots, v_{n}</code></td><td>$v_{1}, v_{2}, \ldots, v_{n}$</td></tr>
<tr><td>rph8-g15q_F00150</td><td>008</td><td>—</td><td><code>s</code></td><td>s</td></tr>
<tr><td>rph8-g15q_F00151</td><td>008</td><td>—</td><td><code>s=\left\{v_{i_{1}}, \ldots, v_{i_{k+1}}\right\}</code></td><td>$s=\left\{v_{i_{1}}, \ldots, v_{i_{k+1}}\right\}$</td></tr>
<tr><td>rph8-g15q_F00152</td><td>008</td><td>—</td><td><code>i_{1}<\cdots<i_{k+1}</code></td><td>$i_{1}<\cdots<i_{k+1}$</td></tr>
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<tr><td>rph8-g15q_F00154</td><td>008</td><td>—</td><td><code>|s\rangle</code></td><td>$|s\rangle$</td></tr>
<tr><td>rph8-g15q_F00155</td><td>008</td><td>—</td><td><code>i_{1}, \ldots, i_{k+1}</code></td><td>$i_{1}, \ldots, i_{k+1}$</td></tr>
<tr><td>rph8-g15q_F00156</td><td>008</td><td>—</td><td><code>|1\rangle</code></td><td>$|1\rangle$</td></tr>
<tr><td>rph8-g15q_F00157</td><td>008</td><td>—</td><td><code>|0\rangle</code></td><td>$|0\rangle$</td></tr>
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Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00160	008	—	`s=\left\{v_{2}, v_{4}, v_{5}\right\}`	$s=\left\{v_{2}, v_{4}, v_{5}\right\}$
rph8-g15q_F00161	008	—	`|s\rangle=|0101100\rangle`	$\lvert s\rangle=\lvert 0101100\rangle$
rph8-g15q_F00162	008	—	`f`	f
rph8-g15q_F00163	008	—	`\left(\mathcal{X}, d_{\mathcal{X}}\right) \rightarrow\left(\mathcal{Y}, d_{\mathcal{Y}}\right)`	$\left(\mathcal{X}, d_{\mathcal{X}}\right) \rightarrow\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00164	008	—	`\left(\mathcal{Y}, d_{\mathcal{Y}}\right)`	$\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00165	008	—	`f: \mathcal{X} \rightarrow \mathcal{Y}`	$f: \mathcal{X} \rightarrow \mathcal{Y}$
rph8-g15q_F00166	008	—	`d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \leqslant d_{\mathcal{X}} \left(x_{1}, x_{2}\right)`	$d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right) \leqslant d_{\mathcal{X}} \left(x_{1}, x_{2}\right)$
rph8-g15q_F00167	009	—	`d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right)=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)`	$d_{\mathcal{Y}}\left(f\left(x_{1}\right), f\left(x_{2}\right)\right)=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)$
rph8-g15q_F00168	009	—	`f:\left(X, d_{X}\right) \rightarrow\left(\mathcal{Y}, d_{\mathcal{Y}}\right)`	$f:\left(X, d_{X}\right) \rightarrow\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00169	009	—	`Y:=f(X)`	$Y:=f(X)$
rph8-g15q_F00170	009	—	`d_{Y}`	d_{Y}
rph8-g15q_F00171	009	—	`d_{\mathcal{Y}}`	$d_{\mathcal{Y}}$
rph8-g15q_F00172	009	—	`K_{\bullet}\left(Y, d_{Y}\right)`	$K_{\bullet}\left(Y, d_{Y}\right)$
rph8-g15q_F00173	009	—	`H_{\bullet}\left(Y, d_{Y}\right)`	$H_{\bullet}\left(Y, d_{Y}\right)$
rph8-g15q_F00174	009	—	`\operatorname{dgm}_{\bullet}\left(Y, d_{Y}\right)`	$\operatorname{dgm}_{\bullet}\left(Y, d_{Y}\right)$
rph8-g15q_F00175	009	—	`\mathcal{X} \xrightarrow{f} \mathcal{Y}`	$\mathcal{X} \xrightarrow{f} \mathcal{Y}$
rph8-g15q_F00176	009	—	`d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow[0, \infty)`	$d_{\mathcal{X}}: \mathcal{X} \times \mathcal{X} \rightarrow[0, \infty)$
rph8-g15q_F00177	009	—	`\left(\mathcal{X}, d_{\mathcal{X}}\right) \xrightarrow{f}\left(\mathcal{Y}, d_{\mathcal{Y}}\right)`	$\left(\mathcal{X}, d_{\mathcal{X}}\right) \xrightarrow{f}\left(\mathcal{Y}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00178	009	—	`L_{p}`	L_{p}
rph8-g15q_F00179	010	—	`\left(\mathcal{X}, d_{\mathcal{X}_{\text {true }}}\right)`	$\left(\mathcal{X}, d_{\mathcal{X}_{\text {true }}}\right)$
rph8-g15q_F00180	010	—	`d_{\mathcal{X}_{\text {true }}}`	$d_{\mathcal{X}_{\text {true }}}$
rph8-g15q_F00181	010	—	`\sigma`	σ
rph8-g15q_F00182	010	—	`\ell_{1}`	ℓ_{1}
rph8-g15q_F00183	010	—	`\ell_{2}`	ℓ_{2}
rph8-g15q_F00184	010	—	`\rho, \sigma \in \mathscr{S}(\mathcal{H})`	$\rho, \sigma \in \mathscr{S}(\mathcal{H})$
rph8-g15q_F00185	010	—	`d_{\mathcal{H}}`	$d_{\mathcal{H}}$
rph8-g15q_F00186	010	—	`\Theta`	Θ
rph8-g15q_F00187	010	—	`\rho: \mathcal{X} \times \Theta \rightarrow \mathscr{S}(\mathcal{H})`	$\rho: \mathcal{X} \times \Theta \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00188	010	—	`\theta \in \Theta`	$\theta \in \Theta$
rph8-g15q_F00189	010	—	`\mathcal{X}^{\prime}`	$\mathcal{X}^{\prime}$
rph8-g15q_F00190	010	—	`\mathscr{S}(\mathcal{H})^{\prime}`	$\mathscr{S}(\mathcal{H})^{\prime}$
rph8-g15q_F00191	010	—	`\rho^{\prime}: \mathcal{X}^{\prime} \rightarrow \mathscr{S}(\mathcal{H})^{\prime}`	$\rho^{\prime}: \mathcal{X}^{\prime} \rightarrow \mathscr{S}(\mathcal{H})^{\prime}$
rph8-g15q_F00192	010	—	`\rho^{\prime}`	$\rho^{\prime}$
rph8-g15q_F00193	011	—	`\mathcal{X}=[-2,2]`	$\mathcal{X}=[-2,2]$
rph8-g15q_F00194	011	—	`\rho:[-2,2] \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)`	$\rho:[-2,2] \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00195	011	—	`\mathscr{S}\left(\mathbb{C}^{2}\right)`	$\mathscr{S}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00196	011	—	`\theta_{1}=0.31, \theta_{2}=1.48, \theta_{3}=0`	$\theta_{1}=0.31, \theta_{2}=1.48, \theta_{3}=0$
rph8-g15q_F00197	011	—	`\theta_{4}=0`	$\theta_{4}=0$
rph8-g15q_F00198	011	—	`\rho: \mathcal{X} \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)`	$\rho: \mathcal{X} \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00199	011	—	`f:`	$f:$
rph8-g15q_F00200	011	—	`\mathcal{X} \rightarrow \mathcal{Y}`	$\mathcal{X} \rightarrow \mathcal{Y}$
rph8-g15q_F00201	011	—	`g: \mathcal{Y} \rightarrow \mathcal{Z}`	$g: \mathcal{Y} \rightarrow \mathcal{Z}$
rph8-g15q_F00202	011	—	`g \circ f: \mathcal{X} \rightarrow \mathcal{Z}`	$g \circ f: \mathcal{X} \rightarrow \mathcal{Z}$
rph8-g15q_F00203	011	—	`f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}`	$f: \mathcal{X} \rightarrow \mathcal{Y}, g: \mathcal{Y} \rightarrow \mathcal{Z}$
rph8-g15q_F00204	011	—	`h: \mathcal{Z} \rightarrow \mathcal{W}`	$h: \mathcal{Z} \rightarrow \mathcal{W}$
rph8-g15q_F00205	011	—	`h \circ(g \circ f)`	$h \circ(g \circ f)$
rph8-g15q_F00206	011	—	`(h \circ g) \circ f`	$(h \circ g) \circ f$
rph8-g15q_F00207	011	—	`\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}`	$\operatorname{id}_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X}$
rph8-g15q_F00208	011	—	`\operatorname{id}_{\mathcal{X}}(x)=x`	$\operatorname{id}_{\mathcal{X}}(x)=x$
rph8-g15q_F00209	011	—	`\operatorname{id}_{\mathcal{Y}} \circ f=f \circ \mathrm{id}_{\mathcal{X}}=f`	$\operatorname{id}_{\mathcal{Y}} \circ f=f \circ \mathrm{id}_{\mathcal{X}}=f$
rph8-g15q_F00210	011	—	`\mathbf{C}`	$\mathbf{C}$
rph8-g15q_F00211		—	`\mathcal{X}, \mathcal{Y}, \ldots`	$\mathcal{X}, \mathcal{Y}, \ldots$
rph8-g15q_F00212	011	—	`\mathcal{X}, \mathcal{Y}`	$\mathcal{X}, \mathcal{Y}$
rph8-g15q_F00213	011	—	`f: \mathcal{X} \rightarrow \mathcal{Y}, \quad g: \mathcal{Y} \rightarrow \mathcal{Z}, \quad`	$f: \mathcal{X} \rightarrow \mathcal{Y}, \quad g: \mathcal{Y} \rightarrow \mathcal{Z}, \quad$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00214	012	—	`f \circ \operatorname{id}_{\mathcal{X}}=`	$f \circ \operatorname{id}_{\mathcal{X}}=$
rph8-g15q_F00215	012	—	`\operatorname{id}_{\mathcal{Y}} \circ f=f`	$\operatorname{id}_{\mathcal{Y}} \circ f=f$
rph8-g15q_F00216	012	—	`\operatorname{id}_{\mathcal{X}}`	$\operatorname{id}_{\mathcal{X}}$
rph8-g15q_F00217	012	—	`{ }^{\text {emb }}`	${ }^{\text {emb }}$
rph8-g15q_F00218	012	—	`{ }^{\mathrm{nif}}`	${ }^{\mathrm{nif}}$
rph8-g15q_F00219	012	—	`\mathbf{M e t}^{\mathrm{nif}}`	$\mathbf{M e t}^{\mathrm{nif}}$
rph8-g15q_F00220	012	—	`{ }^{\mathrm{inf}}`	${ }^{\mathrm{inf}}$
rph8-g15q_F00221	012	—	`{ }^{\text {nif }}`	${ }^{\text {nif }}$
rph8-g15q_F00222	012	—	`\rho:(\mathcal{X}, \alpha) \rightarrow\left(\mathscr{S}(\mathcal{H}), \operatorname{Ad}_{V}\right)`	$\rho:(\mathcal{X}, \alpha) \rightarrow\left(\mathscr{S}(\mathcal{H}), \operatorname{Ad}_{V}\right)$
rph8-g15q_F00223	012	—	`F: G`	$F: G$
rph8-g15q_F00224	013	—	`F(\mathcal{X}, \alpha)=\mathcal{X}`	$F(\mathcal{X}, \alpha)=\mathcal{X}$
rph8-g15q_F00225	013	—	`F(f)=f`	$F(f)=f$
rph8-g15q_F00226	013	—	`F`	F
rph8-g15q_F00227	013	—	`g`	g
rph8-g15q_F00228	013	—	`x \mapsto x`	$x \mapsto x$
rph8-g15q_F00229	013	—	`F: \mathbf{C} \rightarrow \mathbf{D}`	$F: \mathbf{C} \rightarrow \mathbf{D}$
rph8-g15q_F00230		—	`F(\mathcal{X})`	$F(\mathcal{X})$
rph8-g15q_F00231	014	—	`\mathbf{D}`	$\mathbf{D}$
rph8-g15q_F00232	013	—	`F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})`	$F(f): F(\mathcal{X}) \rightarrow F(\mathcal{Y})$
rph8-g15q_F00233	013	—	`F(g \circ f)=`	$F(g \circ f)=$
rph8-g15q_F00234	013	—	`F(g) \circ F(f)`	$F(g) \circ F(f)$
rph8-g15q_F00235	013	—	`F\left(\mathrm{id}_{\mathcal{X}}\right)=\operatorname{id}_{F(\mathcal{X})}`	$F\left(\mathrm{id}_{\mathcal{X}}\right)=\operatorname{id}_{F(\mathcal{X})}$
rph8-g15q_F00236	013	—	`\mathbf{C}(\mathcal{X}, \mathcal{Y})`	$\mathbf{C}(\mathcal{X}, \mathcal{Y})$
rph8-g15q_F00237	013	—	`\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))`	$\mathbf{C}(\mathcal{X}, \mathcal{Y}) \rightarrow \mathbf{D}(F(\mathcal{X}), F(\mathcal{Y}))$
rph8-g15q_F00238	013	—	`f \mapsto F(f)`	$f \mapsto F(f)$
rph8-g15q_F00239	013	—	`\mathbf{C} \rightarrow \mathbf{D}`	$\mathbf{C} \rightarrow \mathbf{D}$
rph8-g15q_F00240	014	—	`F\left(\mathcal{X}^{\prime}\right)=\mathcal{X}`	$F\left(\mathcal{X}^{\prime}\right)=\mathcal{X}$
rph8-g15q_F00241	014	—	`F\left(\mathscr{S}(\mathcal{H})^{\prime}\right)=\mathscr{S}(\mathcal{H})`	$F\left(\mathscr{S}(\mathcal{H})^{\prime}\right)=\mathscr{S}(\mathcal{H})$
rph8-g15q_F00242	014	—	`F: \mathbf{C} \rightarrow`	$F: \mathbf{C} \rightarrow$
rph8-g15q_F00243	014	—	`F\left(\rho^{\prime}\right)=\rho`	$F\left(\rho^{\prime}\right)=\rho$
rph8-g15q_F00244	014	—	`\mathcal{X}^{\prime}:=(\mathcal{X}, \alpha)`	$\mathcal{X}^{\prime}:=(\mathcal{X}, \alpha)$
rph8-g15q_F00245	014	—	`G \ni g \mapsto V_{g} \in \mathscr{U}(\mathcal{H})`	$G \ni g \mapsto V_{g} \in \mathscr{U}(\mathcal{H})$
rph8-g15q_F00246	014	—	`G \ni g \mapsto \operatorname{Ad}_{V_{g}}`	$G \ni g \mapsto \operatorname{Ad}_{V_{g}}$
rph8-g15q_F00247	014	—	`\mathscr{S}(\mathcal{H})^{\prime}:=`	$\mathscr{S}(\mathcal{H})^{\prime}:=$
rph8-g15q_F00248	014	—	`\rho^{\prime}: \mathcal{X}^{\prime} \rightarrow`	$\rho^{\prime}: \mathcal{X}^{\prime} \rightarrow$
rph8-g15q_F00249	014	—	`U`	U
rph8-g15q_F00250	014	—	`\mathbb{R}^{d} \rightarrow \mathscr{U}(\mathcal{H})`	$\mathbb{R}^{d} \rightarrow \mathscr{U}(\mathcal{H})$
rph8-g15q_F00251	014	—	`\mathcal{L}: \mathbb{R}^{d} \rightarrow \mathcal{B}(\mathcal{H})`	$\mathcal{L}: \mathbb{R}^{d} \rightarrow \mathcal{B}(\mathcal{H})$
rph8-g15q_F00252	014	—	`x \in \mathbb{R}^{d}`	$x \in \mathbb{R}^{d}$
rph8-g15q_F00253	014	—	`\mathcal{B}(\mathcal{H})`	$\mathcal{B}(\mathcal{H})$
rph8-g15q_F00254	014	—	`U: \mathbb{R}^{d} \rightarrow \mathscr{U}(\mathcal{H})`	$U: \mathbb{R}^{d} \rightarrow \mathscr{U}(\mathcal{H})$
rph8-g15q_F00255	015	—	`\rho: X \rightarrow \mathscr{S}(\mathcal{H})`	$\rho: X \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00256	015	—	`\rho_{\text {opt }}`	$\rho_{\text {opt }}$
rph8-g15q_F00257	016	—	`\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})`	$\mathcal{L}: \mathcal{X} \rightarrow \mathcal{B}(\mathcal{H})$
rph8-g15q_F00258	016	—	`\rho(x)=U(x)\left|\psi_{0}\right\rangle\left\langle\psi_{0}\right| U(x)^{\dagger}`	$\rho(x)=U(x)\left\lvert\psi_{0}\right\rangle\left\langle\psi_{0}\right\rvert U(x)^{\dagger}$
rph8-g15q_F00259	016	—	`\gamma: \mathbb{R} \rightarrow G`	$\gamma: \mathbb{R} \rightarrow G$
rph8-g15q_F00260	016	—	`\gamma`	γ
rph8-g15q_F00261		—	`\gamma(0)=1`	$\gamma(0)=1$
rph8-g15q_F00262		—	`\gamma(s+t)=\gamma(s) \gamma(t)`	$\gamma(s+t)=\gamma(s) \gamma(t)$
rph8-g15q_F00263		—	`s, t \in \mathbb{R}`	$s, t \in \mathbb{R}$
rph8-g15q_F00264	016	—	`M \in \mathfrak{g}`	$M \in \mathfrak{g}$
rph8-g15q_F00265	016	—	`\mathfrak{g}`	$\mathfrak{g}$
rph8-g15q_F00266	016	—	`\exp : \mathfrak{g} \rightarrow G`	$\exp : \mathfrak{g} \rightarrow G$
rph8-g15q_F00267	016	—	`t \in \mathbb{R}`	$t \in \mathbb{R}$
rph8-g15q_F00268	016	—	`t=0`	$t=0$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00269	016	—	M=\left.\frac{d}{d t} \gamma\right|_{t=0}	$M=\left.\frac{d}{d t} \gamma\right\vert_{t=0}$
rph8-g15q_F00270	016	—	\mathcal{X}=\mathbb{R}	$\mathcal{X}=\mathbb{R}$
rph8-g15q_F00271	016	—	x \mapsto e^{-i x L}	$x \mapsto e^{-i x L}$
rph8-g15q_F00272	016	—	L \in \mathfrak{s u}\left(2^{n}\right)	$L \in \mathfrak{s u}\left(2^{n}\right)$
rph8-g15q_F00273	016	—	\exp : \mathbb{R}^{d} \rightarrow \mathbb{R}^{d}	$\exp : \mathbb{R}^{d} \rightarrow \mathbb{R}^{d}$
rph8-g15q_F00274	016	—	\exp (x)=x	$\exp (x)=x$
rph8-g15q_F00275	016	—	[V, W]:=V W-W V	$[V, W]:=V W-W V$
rph8-g15q_F00276	016	—	H	H
rph8-g15q_F00277	016	—	f: G \rightarrow H	$f: G \rightarrow H$
rph8-g15q_F00278	016	—	\mathcal{M}: \mathfrak{g} \rightarrow \mathfrak{h}	$\mathcal{M}: \mathfrak{g} \rightarrow \mathfrak{h}$
rph8-g15q_F00279	016	—	W \in \mathfrak{g}	$W \in \mathfrak{g}$
rph8-g15q_F00280	016	—	\mathcal{M}	$\mathcal{M}$
rph8-g15q_F00281	016	—	W_{1}, W_{2} \in \mathfrak{g}	$W_{1}, W_{2} \in \mathfrak{g}$
rph8-g15q_F00282	017	—	U: \mathcal{X} \rightarrow \mathscr{U}\left(\mathbb{C}^{2}\right)	$U: \mathcal{X} \rightarrow \mathscr{U}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00283	017	—	(s, t) \in \mathbb{R}^{2}	$(s, t) \in \mathbb{R}^{2}$
rph8-g15q_F00284	017	—	\mathcal{M}: \mathbb{R}^{2} \rightarrow \mathfrak{s u}(2)	$\mathcal{M}: \mathbb{R}^{2} \rightarrow \mathfrak{s u}(2)$
rph8-g15q_F00285	017	—	\mathcal{M}((1,0))=-i X	$\mathcal{M}((1,0))=-i X$
rph8-g15q_F00286	017	—	\mathcal{M}((0,1))=-i Y	$\mathcal{M}((0,1))=-i Y$
rph8-g15q_F00287	017	—	\mathcal{M}((s, t))=-i s X-i t Y	$\mathcal{M}((s, t))=-i s X-i t Y$
rph8-g15q_F00288	017	—	\exp : \mathbb{R}^{2} \rightarrow \mathbb{R}^{2}	$\exp : \mathbb{R}^{2} \rightarrow \mathbb{R}^{2}$
rph8-g15q_F00289	017	—	\mathbb{R}^{2}	$\mathbb{R}^{2}$
rph8-g15q_F00290	017	—	\mathbb{R}_{\alpha} \subset \mathbb{R}^{2}	$\mathbb{R}_{\alpha} \subset \mathbb{R}^{2}$
rph8-g15q_F00291	017	—	\left.U\right|_{\mathbb{R}_{\alpha}}: \mathbb{R}_{\alpha} \rightarrow \mathscr{U}\left(\mathbb{C}^{2}\right)	$\left.U\right\vert_{\mathbb{R}_{\alpha}}: \mathbb{R}_{\alpha} \rightarrow \mathscr{U}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00292	017	—	W:=\alpha \cdot(X, Y)	$W:=\alpha \cdot(X, Y)$
rph8-g15q_F00293	017	—	X, Y	X, Y
rph8-g15q_F00294	017	—	W	W
rph8-g15q_F00295	017	—	\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2}	$\mathbb{R} \xrightarrow{\alpha} \mathbb{R}^{2}$
rph8-g15q_F00296	017	—	r \in \mathbb{R}	$r \in \mathbb{R}$
rph8-g15q_F00297	017	—	\alpha(r):=r \alpha	$\alpha(r):=r \alpha$
rph8-g15q_F00298	017	—	\mathscr{U}\left(\mathbb{C}^{2}\right)	$\mathscr{U}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00299	017	—	U(r \alpha)	$U(r \alpha)$
rph8-g15q_F00300	017	—	f: G \rightarrow	$f: G \rightarrow$
rph8-g15q_F00301	017	—	\mathbb{R} \xrightarrow{\gamma} G	$\mathbb{R} \xrightarrow{\gamma} G$
rph8-g15q_F00302	017	—	\mathbb{R} \xrightarrow{\gamma} G \xrightarrow{f} H	$\mathbb{R} \xrightarrow{\gamma} G \xrightarrow{f} H$
rph8-g15q_F00303	017	—	G \xrightarrow{f} H	$G \xrightarrow{f} H$
rph8-g15q_F00304	017	—	x \in \mathfrak{g}	$x \in \mathfrak{g}$
rph8-g15q_F00305	017	—	1_{G} \in G	$1_{G} \in G$
rph8-g15q_F00306	017	—	\mathcal{M}:=	$\mathcal{M}:=$
rph8-g15q_F00307	017	—	\left.D f\right|_{1_{G}}	$\left.D f\right\vert_{1_{G}}$
rph8-g15q_F00308	017	—	\gamma(0)=1_{G}	$\gamma(0)=1_{G}$
rph8-g15q_F00309	017	—	\left.\frac{d}{d t} \gamma\right|_{t=0}=x \in \mathfrak{g}	$\left.\frac{d}{d t} \gamma\right\vert_{t=0}=x \in \mathfrak{g}$
rph8-g15q_F00310	017	—	\left.\frac{d}{d t}(f \circ \gamma)\right|_{t=0}=\mathcal{M}(x)	$\left.\frac{d}{d t}(f \circ \gamma)\right\vert_{t=0}=\mathcal{M}(x)$
rph8-g15q_F00311	017	—	\left.\frac{d}{d t} \gamma\right|_{t=0}=x	$\left.\frac{d}{d t} \gamma\right\vert_{t=0}=x$
rph8-g15q_F00312	017	—	f \circ \gamma	$f \circ \gamma$
rph8-g15q_F00313	017	—	y \in \mathfrak{h}	$y \in \mathfrak{h}$
rph8-g15q_F00314	018	—	G=\mathbb{R}^{d}	$G=\mathbb{R}^{d}$
rph8-g15q_F00315	018	—	\mathfrak{g}=\mathbb{R}^{d}	$\mathfrak{g}=\mathbb{R}^{d}$
rph8-g15q_F00316	018	—	\mathbb{R}^{d} \xrightarrow{f} G	$\mathbb{R}^{d} \xrightarrow{f} G$
rph8-g15q_F00317	018	—	\mathcal{M}: \mathbb{R}^{d} \rightarrow	$\mathcal{M}: \mathbb{R}^{d} \rightarrow$
rph8-g15q_F00318	018	—	f: \mathbb{R}^{d} \rightarrow G	$f: \mathbb{R}^{d} \rightarrow G$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00319	018	—	`\mathbb{R}^{d} \rightarrow \mathscr{U}\left(\mathbb{C}^{n}\right)`	$\mathbb{R}^{d} \rightarrow \mathscr{U}\left(\mathbb{C}^{n}\right)$
rph8-g15q_F00320	018	—	`\mathcal{L}: \mathbb{R}^{d} \rightarrow \mathcal{B}\left(\mathbb{C}^{n}\right)`	$\mathcal{L}: \mathbb{R}^{d} \rightarrow \mathcal{B}\left(\mathbb{C}^{n}\right)$
rph8-g15q_F00321	018	—	`\mathbb{R}^{d} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{n}\right)`	$\mathbb{R}^{d} \xrightarrow{U} \mathscr{U}\left(\mathbb{C}^{n}\right)$
rph8-g15q_F00322	018	—	`U(x+y)=e^{-i \mathcal{L}(x+y)}`	$U(x+y)=e^{-i \mathcal{L}(x+y)}$
rph8-g15q_F00323	018	—	`U(x) U(y)=e^{-i \mathcal{L}(x)} e^{-i \mathcal{L}(y)}`	$U(x) U(y)=e^{-i \mathcal{L}(x)} e^{-i \mathcal{L}(y)}$
rph8-g15q_F00324	018	—	`X \subseteq \mathbb{R}^{2}`	$X \subseteq \mathbb{R}^{2}$
rph8-g15q_F00325	018	—	`c: \mathcal{X} \rightarrow\{-1,0,+1\}`	$c: \mathcal{X} \rightarrow\{-1,0,+1\}$
rph8-g15q_F00326	018	—	`O \in`	$O \in$
rph8-g15q_F00327	018	—	`O`	O
rph8-g15q_F00328	018	—	`\{-1,+1\}`	$\{-1,+1\}$
rph8-g15q_F00329	018	—	`\rho(x)`	$\rho(x)$
rph8-g15q_F00330	018	—	`y`	y
rph8-g15q_F00331	018	—	`y^{-1}(0) \subseteq \mathcal{X}`	$y^{-1}(0) \subseteq \mathcal{X}$
rph8-g15q_F00332	018	—	`x_{1}, x_{2} \in \mathbb{R}`	$x_{1}, x_{2} \in \mathbb{R}$
rph8-g15q_F00333	018	—	`\left|\psi_{0}\right\rangle \in \mathcal{H}=\mathbb{C}^{4}`	$\left\|\psi_{0}\right\rangle \in \mathcal{H}=\mathbb{C}^{4}$
rph8-g15q_F00334	018	—	`p=0.99`	$p=0.99$
rph8-g15q_F00335	018	—	`\left(x_{1}, x_{2}\right)`	$\left(x_{1}, x_{2}\right)$
rph8-g15q_F00336	018	—	`y\left(x_{1}, x_{2}\right)<0`	$y\left(x_{1}, x_{2}\right)<0$
rph8-g15q_F00337	018	—	`y\left(x_{1}, x_{2}\right)>0`	$y\left(x_{1}, x_{2}\right)>0$
rph8-g15q_F00338	018	—	`y\left(x_{1}, x_{2}\right)=`	$y\left(x_{1}, x_{2}\right)=$
rph8-g15q_F00339	018	—	`U: \mathbb{R}^{2} \rightarrow \mathscr{U}\left(\mathbb{C}^{4}\right)`	$U: \mathbb{R}^{2} \rightarrow \mathscr{U}\left(\mathbb{C}^{4}\right)$
rph8-g15q_F00340	018	—	`U(x)=e^{-i \mathcal{L}(x)}`	$U(x)=e^{-i \mathcal{L}(x)}$
rph8-g15q_F00341	018	—	`\mathcal{L}: \mathbb{R}^{2} \rightarrow \mathfrak{s u}(4)`	$\mathcal{L}: \mathbb{R}^{2} \rightarrow \mathfrak{s u}(4)$
rph8-g15q_F00342	018	—	`e_{1}, e_{2}`	e_{1}, e_{2}
rph8-g15q_F00343	018	—	`L_{1}:=\mathcal{L}\left(e_{1}\right)`	$L_{1}:=\mathcal{L}\left(e_{1}\right)$
rph8-g15q_F00344	018	—	`L_{2}:=\mathcal{L}\left(e_{2}\right)`	$L_{2}:=\mathcal{L}\left(e_{2}\right)$
rph8-g15q_F00345	018	—	`a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}`	$a_{11}, a_{22}, a_{33}, a_{44} \in \mathbb{R}, a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34} \in \mathbb{C}$
rph8-g15q_F00346	018	—	`a_{11}+a_{22}+a_{33}+a_{44}=0`	$a_{11}+a_{22}+a_{33}+a_{44}=0$
rph8-g15q_F00347	018	—	`L_{2}`	L_{2}
rph8-g15q_F00348	018	—	`2(4+2(6)-1)=30`	$2(4+2(6)-1)=30$
rph8-g15q_F00349	018	—	`\mathbb{R}^{2} \rightarrow \mathscr{U}\left(\mathbb{C}^{4}\right)`	$\mathbb{R}^{2} \rightarrow \mathscr{U}\left(\mathbb{C}^{4}\right)$
rph8-g15q_F00350	018	—	`L_{2}=`	$L_{2}=$
rph8-g15q_F00351	019	—	`\left\{X \otimes X, L_{1}\right\}=0`	$\left\{X \otimes X, L_{1}\right\}=0$
rph8-g15q_F00352	019	—	`\mathbb{R}^{n}`	$\mathbb{R}^{n}$
rph8-g15q_F00353	019	—	`\mathcal{X}=\mathbb{R}^{n}`	$\mathcal{X}=\mathbb{R}^{n}$
rph8-g15q_F00354	019	—	`\left(\mathcal{Y}=\mathbb{R}^{m}, d_{\mathcal{Y}}\right)`	$\left(\mathcal{Y}=\mathbb{R}^{m}, d_{\mathcal{Y}}\right)$
rph8-g15q_F00355	019	—	`\mathbb{R}^{n} \xrightarrow{\varphi} \mathbb{R}^{m}`	$\mathbb{R}^{n} \xrightarrow{\varphi} \mathbb{R}^{m}$
rph8-g15q_F00356	019	—	`m \times n`	$m \times n$
rph8-g15q_F00357	019	—	`V`	V
rph8-g15q_F00358	019	—	`V^{T}`	V^{T}
rph8-g15q_F00359	019	—	`\left(\mathbb{R}^{m}, A\right)`	$\left(\mathbb{R}^{m}, A\right)$
rph8-g15q_F00360	019	—	`m \in \mathbb{N}`	$m \in \mathbb{N}$
rph8-g15q_F00361	019	—	`A`	A
rph8-g15q_F00362	019	—	`m \times m`	$m \times m$
rph8-g15q_F00363	019	—	`\mathbb{R}^{m}, A`	$\mathbb{R}^{m}, A$
rph8-g15q_F00364	019	—	`\mathbb{R}^{n}, B`	$\mathbb{R}^{n}, B$
rph8-g15q_F00365	019	—	`n \times m`	$n \times m$
rph8-g15q_F00366	019	—	`V^{T} B V=A`	$V^{T} B V=A$
rph8-g15q_F00367	019	—	`\mathbb{R}^{m}`	$\mathbb{R}^{m}$
rph8-g15q_F00368	019	—	`\mathbb{R}^{m} \xrightarrow{V} \mathbb{R}^{n}`	$\mathbb{R}^{m} \xrightarrow{V} \mathbb{R}^{n}$
rph8-g15q_F00369	019	—	`y \in \mathcal{Y}`	$y \in \mathcal{Y}$
rph8-g15q_F00370	019	—	`x_{3} \in \mathcal{X}`	$x_{3} \in \mathcal{X}$
rph8-g15q_F00371	019	—	`d_{\mathcal{X}}(x, x)=0`	$d_{\mathcal{X}}(x, x)=0$
rph8-g15q_F00372	019	—	`f\left(x_{1}\right)=f\left(x_{2}\right)`	$f\left(x_{1}\right)=f\left(x_{2}\right)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00373	019	—	`x_{1}, x_{2}`	x_{1}, x_{2}
rph8-g15q_F00374	019	—	`X \subset \mathcal{X}`	$X \subset \mathcal{X}$
rph8-g15q_F00375	020	—	`\{A, B\}`	$\{A, B\}$
rph8-g15q_F00376	020	—	`B`	B
rph8-g15q_F00377	020	—	`c: \mathcal{X} \rightarrow\{-1,0,1\}`	$c: \mathcal{X} \rightarrow\{-1,0,1\}$
rph8-g15q_F00378	020	—	`\# A`	$\# A$
rph8-g15q_F00379	020	—	`\# B`	$\# B$
rph8-g15q_F00380	020	—	`\mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})`	$\mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00381	020	—	`\rho: \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})`	$\rho: \Theta \times \mathcal{X} \rightarrow \mathscr{S}(\mathcal{H})$
rph8-g15q_F00382	020	—	`\Theta=[0,2 \pi)^{4}`	$\Theta=[0,2 \pi)^{4}$
rph8-g15q_F00383	020	—	`\theta=\left(\theta_{1}, \theta_{2}, \theta_{3}, \theta_{4}\right) \in \Theta`	$\theta=\left(\theta_{1}, \theta_{2}, \theta_{3}, \theta_{4}\right) \in \Theta$
rph8-g15q_F00384	020	—	`\rho_{\theta}(x):=\rho(\theta, x)`	$\rho_{\theta}(x):=\rho(\theta, x)$
rph8-g15q_F00385	020	—	`U: \Theta \times \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})`	$U: \Theta \times \mathcal{X} \rightarrow \mathscr{U}(\mathcal{H})$
rph8-g15q_F00386	020	—	`\rho_{A}`	ρ_{A}
rph8-g15q_F00387	020	—	`\rho_{B}`	ρ_{B}
rph8-g15q_F00388	020	—	`\lambda \in(0,1]`	$\lambda \in(0,1]$
rph8-g15q_F00389	020	—	`\rho_{A}-\rho_{B}`	$\rho_{A}-\rho_{B}$
rph8-g15q_F00390	020	—	`\{-\lambda,+\lambda\}`	$\{-\lambda,+\lambda\}$
rph8-g15q_F00391	020	—	`\lambda_{1}`	λ_{1}
rph8-g15q_F00392	020	—	`\lambda_{2}`	λ_{2}
rph8-g15q_F00393	020	—	`\rho_{A} \neq \rho_{B}`	$\rho_{A} \neq \rho_{B}$
rph8-g15q_F00394	020	—	`\lambda_{1}+\lambda_{2}=0`	$\lambda_{1}+\lambda_{2}=0$
rph8-g15q_F00395	020	—	`\lambda_{2}=-\lambda_{1}`	$\lambda_{2}=-\lambda_{1}$
rph8-g15q_F00396	020	—	`\lambda=\left|\lambda_{1}\right|`	$\lambda=\left\lvert\lambda_{1}\right\rvert$
rph8-g15q_F00397	020	—	`A \cup B=X`	$A \cup B=X$
rph8-g15q_F00398	020	—	`O=\rho_{A}-\rho_{B}`	$O=\rho_{A}-\rho_{B}$
rph8-g15q_F00399	020	—	`\lambda>0`	$\lambda>0$
rph8-g15q_F00400	020	—	`c`	c
rph8-g15q_F00401	020	—	`x \in X`	$x \in X$
rph8-g15q_F00402	020	—	`\operatorname{sgn}:(-\infty, 0) \cup(0, \infty) \rightarrow\{-1,+1\}`	$\operatorname{sgn}:(-\infty, 0) \cup(0, \infty) \rightarrow\{-1,+1\}$
rph8-g15q_F00403	020	—	`\operatorname{sgn}(x)=\frac{x}{|x|}`	$\operatorname{sgn}(x)=\frac{x}{\lvert x\rvert}$
rph8-g15q_F00404	020	—	`\Pi_{+}-\Pi_{-}`	$\Pi_{+}-\Pi_{-}$
rph8-g15q_F00405	020	—	`\Pi_{+}`	Π_{+}
rph8-g15q_F00406	020	—	`\Pi_{-}`	Π_{-}
rph8-g15q_F00407	020	—	`\theta_{1}, \theta_{2}, \theta_{3}`	$\theta_{1}, \theta_{2}, \theta_{3}$
rph8-g15q_F00408	020	—	`\theta_{4}`	θ_{4}
rph8-g15q_F00409	020	—	`\rho: \Theta \times \mathcal{X} \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)`	$\rho: \Theta \times \mathcal{X} \rightarrow \mathscr{S}\left(\mathbb{C}^{2}\right)$
rph8-g15q_F00410	020	—	`\mathcal{X} \times \mathcal{X}`	$\mathcal{X} \times \mathcal{X}$
rph8-g15q_F00411	021	—	`G: \mathbf{D} \rightarrow \mathbf{E}`	$G: \mathbf{D} \rightarrow \mathbf{E}$
rph8-g15q_F00412	021	—	`G \circ F: \mathbf{C} \rightarrow \mathbf{E}`	$G \circ F: \mathbf{C} \rightarrow \mathbf{E}$
rph8-g15q_F00413	021	—	`G(F(\mathcal{X}))`	$G(F(\mathcal{X}))$
rph8-g15q_F00414	021	—	`G(F(f))`	$G(F(f))$
rph8-g15q_F00415	021	—	`F, G`	F, G
rph8-g15q_F00416	021	—	`\eta`	η
rph8-g15q_F00417	021	—	`\eta: F \Rightarrow G`	$\eta: F \Rightarrow G$
rph8-g15q_F00418	021	—	`\eta_{\mathcal{X}}: F(\mathcal{X}) \rightarrow G(\mathcal{X})`	$\eta_{\mathcal{X}}: F(\mathcal{X}) \rightarrow G(\mathcal{X})$
rph8-g15q_F00419	021	—	`G(f) \circ \eta_{\mathcal{X}}=\eta_{\mathcal{Y}} \circ F(f)`	$G(f) \circ \eta_{\mathcal{X}}=\eta_{\mathcal{Y}} \circ F(f)$
rph8-g15q_F00420	021	—	`\mathcal{C}`	$\mathcal{C}$
rph8-g15q_F00421	021	—	`\mathbb{C}`	$\mathbb{C}$
rph8-g15q_F00422	021	—	`\left(\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}, 0,+, \cdot\right)`	$\left(\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}, 0,+, \cdot\right)$
rph8-g15q_F00423	021	—	`\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}`	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00424	021	—	`v`	v
rph8-g15q_F00425	021	—	`\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}`	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00426	021	—	`v: \mathcal{X} \rightarrow \mathbb{C}`	$v: \mathcal{X} \rightarrow \mathbb{C}$
rph8-g15q_F00427	021	—	`v(x)=0`	$v(x)=0$
rph8-g15q_F00428	021	—	`v, w \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}`	$v, w \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00429	021	—	v+w	$v+w$
rph8-g15q_F00430	021	—	(v+w)(x):=	$(v+w)(x):=$
rph8-g15q_F00431	021	—	v(x)+w(x)	$v(x)+w(x)$
rph8-g15q_F00432	021	—	\lambda \in \mathbb{C}	$\lambda \in \mathbb{C}$
rph8-g15q_F00433	021	—	\lambda \cdot v	$\lambda \cdot v$
rph8-g15q_F00434	021	—	\lambda v	λv
rph8-g15q_F00435	021	—	\lambda v(x)	$\lambda v(x)$
rph8-g15q_F00436	021	—	\delta_{x}	δ_{x}
rph8-g15q_F00437	021	—	\delta_{x}\left(x^{\prime}\right)=\delta_{x x^{\prime}}	$\delta_{x}\left(x^{\prime}\right)=\delta_{x x^{\prime}}$
rph8-g15q_F00438	021	—	x^{\prime}=x	$x^{\prime}=x$
rph8-g15q_F00439	021	—	x^{\prime} \neq	$x^{\prime} \neq$
rph8-g15q_F00440	021	—	|x\rangle	$\lvert x\rangle$
rph8-g15q_F00441	021	—	\left\langle x^{\prime} \mid x\right\rangle=\delta_{x x^{\prime}}	$\left\langle x^{\prime} \mid x\right\rangle=\delta_{x x^{\prime}}$
rph8-g15q_F00442	021	—	\mathcal{X}=\mathbb{Z}_{2}^{n}=	$\mathcal{X}=\mathbb{Z}_{2}^{n}=$
rph8-g15q_F00443	021	—	\{0,1\}^{n}	$\{0,1\}^{n}$
rph8-g15q_F00444	021	—	x=\left(x_{0}, x_{1}, \ldots, x_{n-1}\right)	$x=\left(x_{0}, x_{1}, \ldots, x_{n-1}\right)$
rph8-g15q_F00445	021	—	\left\{0,1,2, \ldots, 2^{n}-1\right\}	$\left\{0,1,2, \ldots, 2^{n}-1\right\}$
rph8-g15q_F00446	021	—	\mathbb{C}^{\mathcal{X}}	$\mathbb{C}^{\mathcal{X}}$
rph8-g15q_F00447	021	—	2^{n}	2^{n}
rph8-g15q_F00448	021	—	L_{f}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow	$L_{f}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow$
rph8-g15q_F00449	021	—	\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}	$\mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00450	021	—	|y\rangle	$\lvert y\rangle$
rph8-g15q_F00451	021	—	L_{f}	L_{f}
rph8-g15q_F00452	021	—	|f(x)\rangle	$\lvert f(x)\rangle$
rph8-g15q_F00453	021	—	\mathbb{C}^{\mathcal{Y}}	$\mathbb{C}^{\mathcal{Y}}$
rph8-g15q_F00454	021	—	L_{f}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}	$L_{f}: \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{Y}}$
rph8-g15q_F00455	021	—	(V, 0,+, \cdot)	$(V, 0,+, \cdot)$
rph8-g15q_F00456	022	—	F(\mathcal{X})=	$F(\mathcal{X})=$
rph8-g15q_F00457	022	—	G(F(\mathcal{X}))=\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}	$G(F(\mathcal{X}))=\mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00458	022	—	G \circ F	$G \circ F$
rph8-g15q_F00459	022	—	\mathrm{id}_{\text {Set }}	$\mathrm{id}_{\text {Set }}$
rph8-g15q_F00460	022	—	\eta: \mathrm{id}_{\text {Set }} \Rightarrow	$\eta: \mathrm{id}_{\text {Set }} \Rightarrow$
rph8-g15q_F00461	022	—	\eta_{\mathcal{X}}: \operatorname{id}_{\text {Set }}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))	$\eta_{\mathcal{X}}: \operatorname{id}_{\text {Set }}(\mathcal{X}) \rightarrow G(F(\mathcal{X}))$
rph8-g15q_F00462	022	—	\eta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}	$\eta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00463	022	—	\eta_{\mathcal{X}}(x)=|x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}	$\eta_{\mathcal{X}}(x)=\lvert x\rangle \in \mathbb{C}_{\mathrm{fs}}^{\mathcal{X}}$
rph8-g15q_F00464	022	—	\{|x\rangle\}	$\{\lvert x\rangle\}$
rph8-g15q_F00465	022	—	\eta_{\mathcal{Y}}(f(x))=L_{f}\left(\eta_{\mathcal{X}}(x)\right)	$\eta_{\mathcal{Y}}(f(x))=L_{f}\left(\eta_{\mathcal{X}}(x)\right)$
rph8-g15q_F00466	022	—	|f(x)\rangle=L_{f}|x\rangle	$\lvert f(x)\rangle=L_{f}\lvert x\rangle$
rph8-g15q_F00467	022	—	\mathbb{B} G	$\mathbb{B} G$
rph8-g15q_F00468	022	—	\alpha: \mathbb{B} G \rightarrow	$\alpha: \mathbb{B} G \rightarrow$
rph8-g15q_F00469	022	—	g^{-1}	g^{-1}
rph8-g15q_F00470	022	—	g g^{-1}=1_{G}=g^{-1} g	$g g^{-1}=1_{G}=g^{-1} g$
rph8-g15q_F00471	022	—	\operatorname{id}_{\mathcal{X}}=\alpha_{e}=\alpha_{g g^{-1}}=\alpha_{g} \circ \alpha_{g^{-1}}	$\operatorname{id}_{\mathcal{X}}=\alpha_{e}=\alpha_{g g^{-1}}=\alpha_{g} \circ \alpha_{g^{-1}}$
rph8-g15q_F00472	022	—	\operatorname{id}_{\mathcal{X}}=\alpha_{g^{-1}} \circ \alpha_{g}	$\operatorname{id}_{\mathcal{X}}=\alpha_{g^{-1}} \circ \alpha_{g}$
rph8-g15q_F00473	022	—	\alpha_{g}	α_{g}
rph8-g15q_F00474	022	—	\alpha_{g^{-1}}	$\alpha_{g^{-1}}$
rph8-g15q_F00475	022	—	\alpha, \beta	α, β
rph8-g15q_F00476	022	—	\mathbb{B} G \rightarrow	$\mathbb{B} G \rightarrow$
rph8-g15q_F00477	022	—	\mathcal{X}:=\alpha(\bullet)	$\mathcal{X}:=\alpha(\bullet)$
rph8-g15q_F00478	022	—	\mathcal{Y}:=\beta(\bullet)	$\mathcal{Y}:=\beta(\bullet)$
rph8-g15q_F00479	022	—	f: \alpha \Rightarrow \beta	$f: \alpha \Rightarrow \beta$
rph8-g15q_F00480	022	—	\bullet	$\bullet$
rph8-g15q_F00481	025	—	w^{*}	w^{*}
rph8-g15q_F00482	025	—	x, y, \ldots \in \mathbb{R}	$x, y, \ldots \in \mathbb{R}$
rph8-g15q_F00483	025	—	x \rightarrow y	$x \rightarrow y$
rph8-g15q_F00484	025	—	x \leqslant y	$x \leqslant y$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

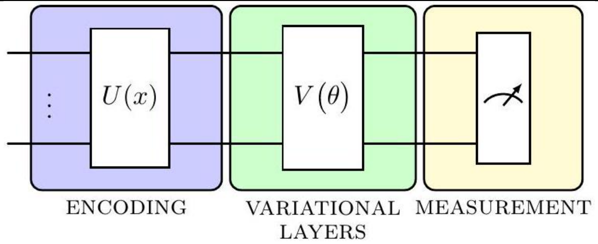
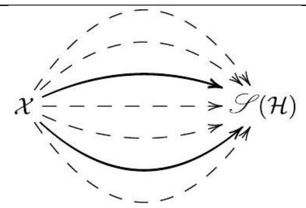
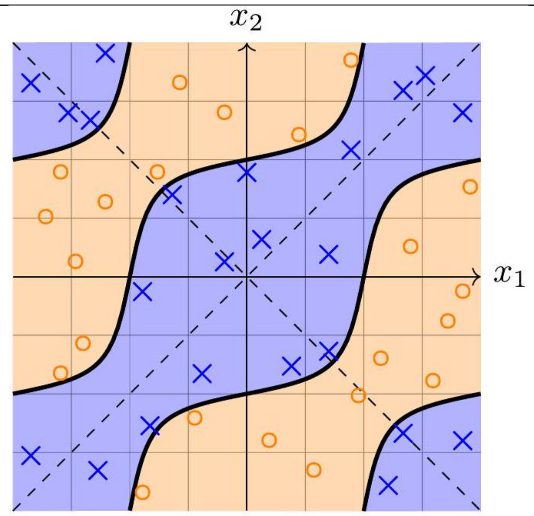
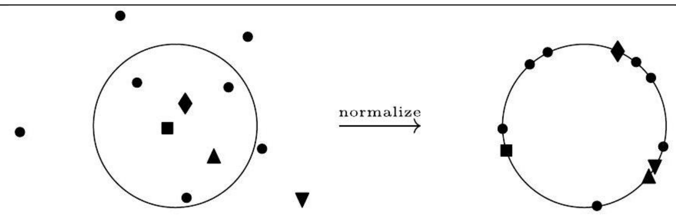
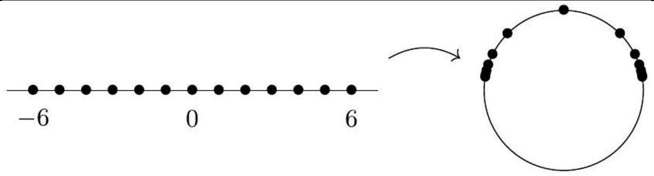
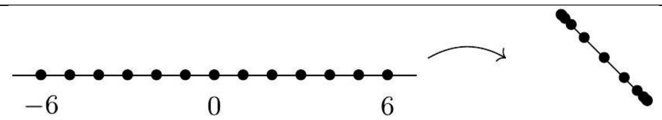
Identifier	Page	Conf.	LaTeX source	Rendered
rph8-g15q_F00485	025	—	`\alpha \in \mathbb{R}^{2}`	$\alpha \in \mathbb{R}^{2}$
rph8-g15q_F00486	025	—	`\alpha: \mathbb{R} \rightarrow \mathbb{R}^{2}`	$\alpha: \mathbb{R} \rightarrow \mathbb{R}^{2}$
rph8-g15q_F00487	025	—	`1 \in \mathbb{R}`	$1 \in \mathbb{R}$
rph8-g15q_F00488	025	—	`\operatorname{Tr}[\sigma O]`	$\operatorname{Tr}[\sigma O]$
rph8-g15q_F00489	025	—	`\{\sigma \in \mathscr{S}(\mathcal{H}): \operatorname{Tr}[\sigma O]=0\}`	$\{\sigma \in \mathscr{S}(\mathcal{H}): \operatorname{Tr}[\sigma O]=0\}$

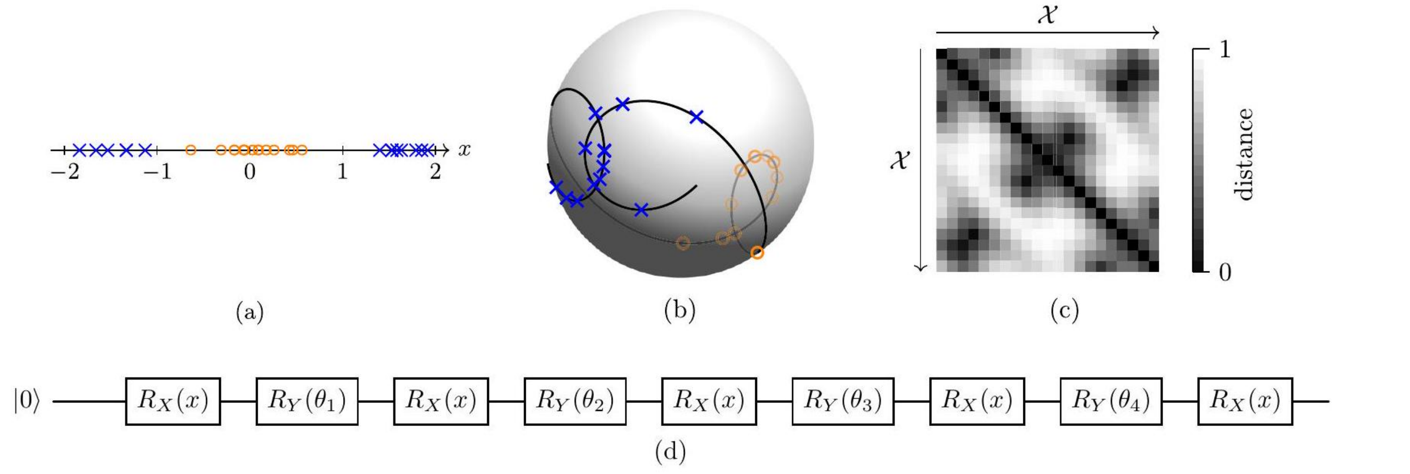
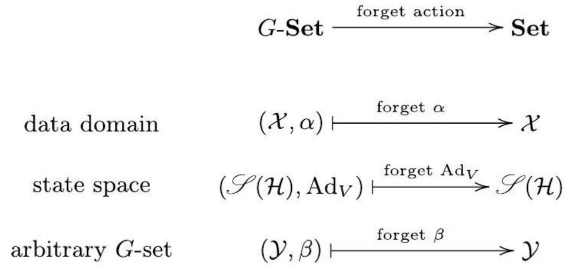
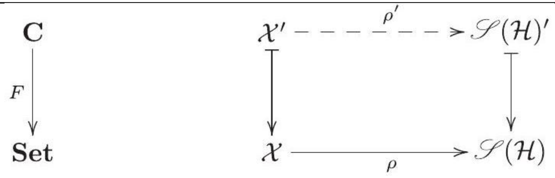
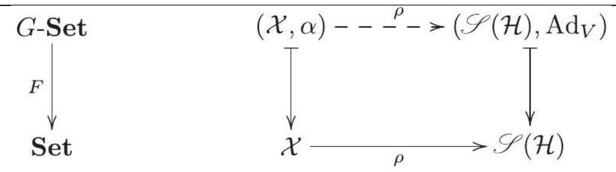
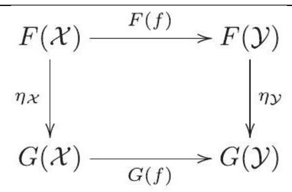
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with `pdfdrill vision` (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
rph8-g15q_DIA_0001	003		tex.zip (local)
rph8-g15q_DIA_0002	004		tex.zip (local)
rph8-g15q_DIA_0003	004		tex.zip (local)
rph8-g15q_DIA_0004	007		tex.zip (local)
rph8-g15q_DIA_0005	007		tex.zip (local)
rph8-g15q_DIA_0006	008		tex.zip (local)
rph8-g15q_DIA_0007	008	$(X, d_X) \longmapsto K_\bullet(X, d_X) \longmapsto H_\bullet(X, d_X) \longmapsto \mathrm{dgm}_\bullet(X, d_X)$ finite metric space — filtered simplicial complex — persistence vector space — persistence diagram	tex.zip (local)

Identifier	Page	Image	Source
rph8-g15q_DIA_0008	011		tex.zip (local)
rph8-g15q_DIA_0009	013		tex.zip (local)
rph8-g15q_DIA_0010	014		tex.zip (local)
rph8-g15q_DIA_0011	014		tex.zip (local)
rph8-g15q_DIA_0012	021		tex.zip (local)